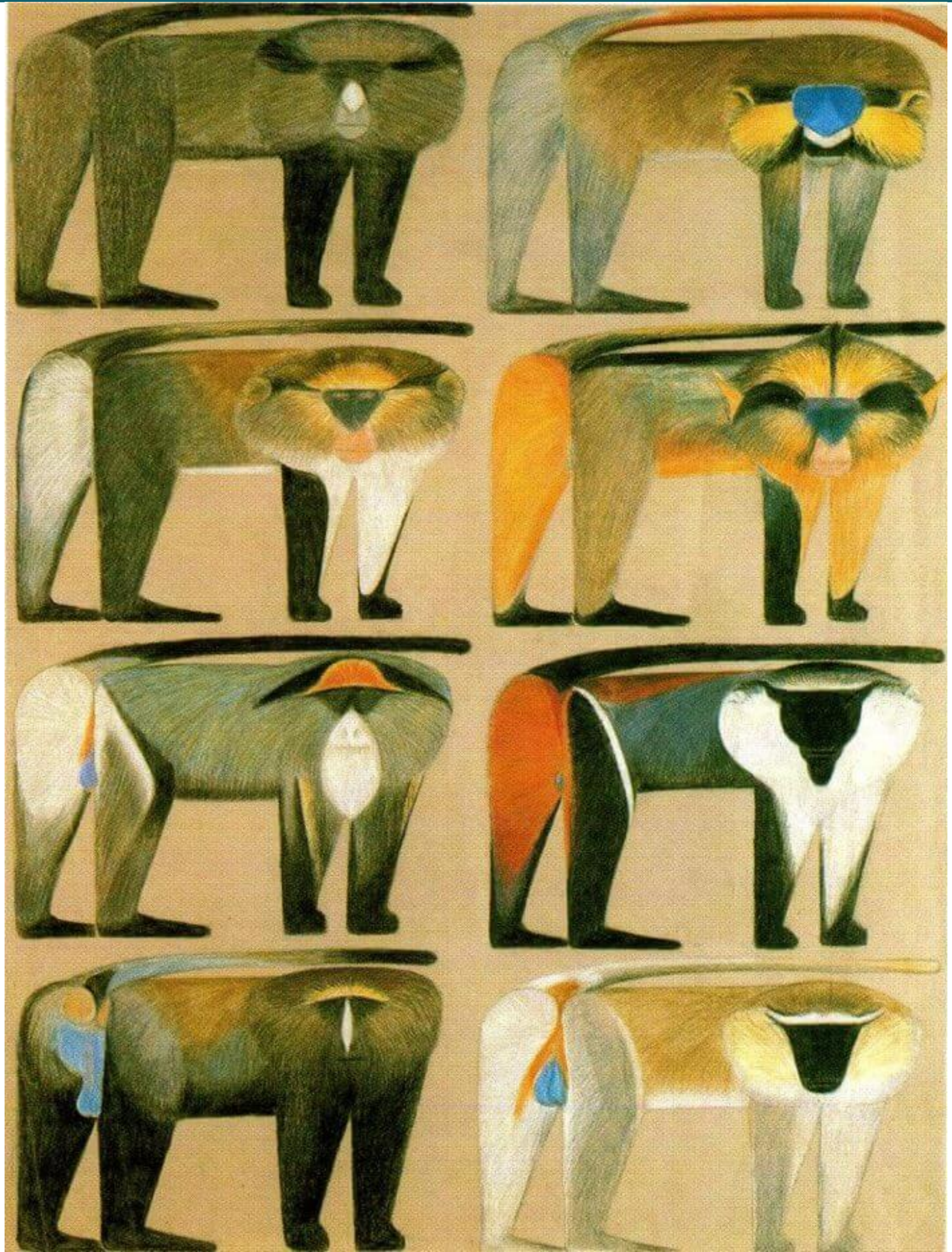


Culture and genetic variation

Dr. Laura van Holstein

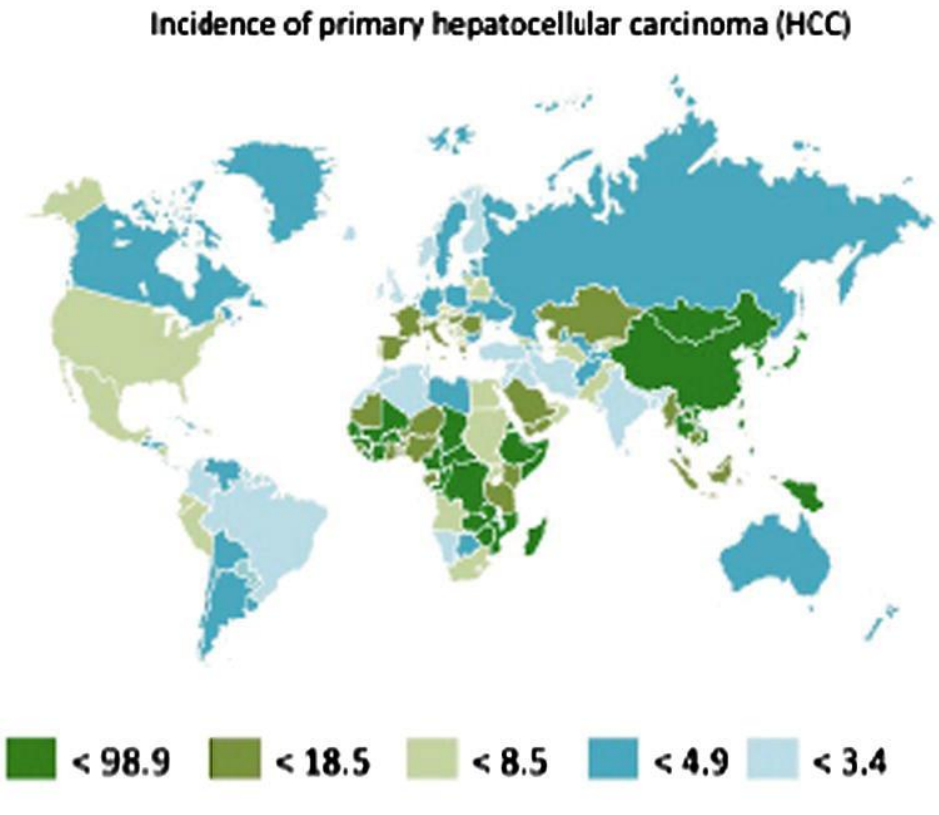
lavanho@emory.edu

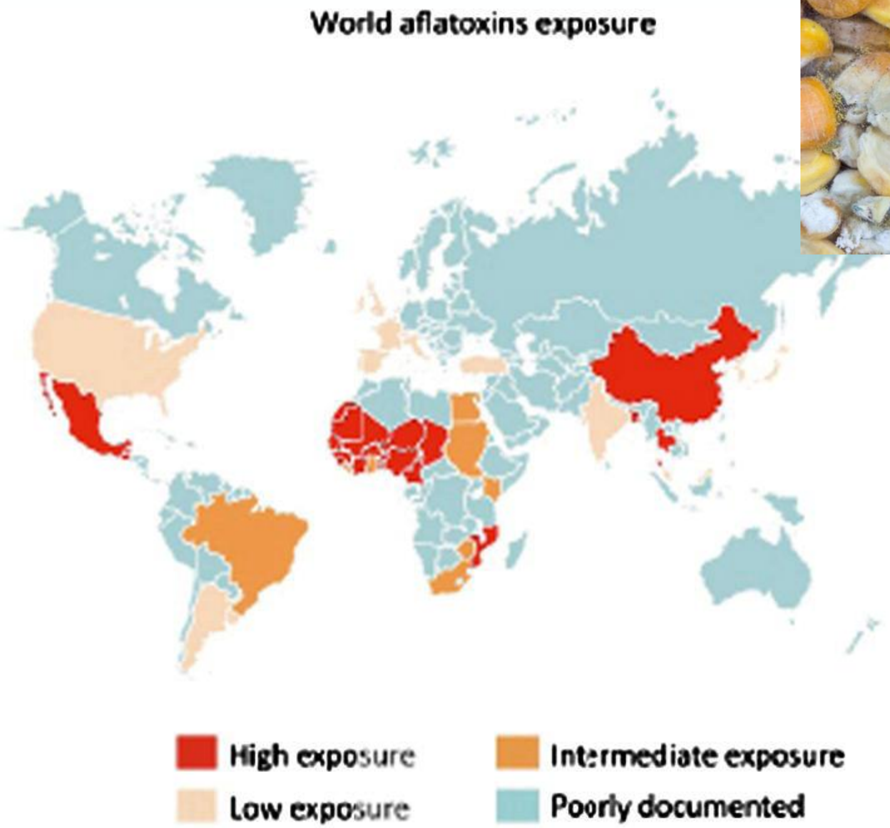
Room 215

Which of these processes might be influenced by **culture** (or behaviour more generally)?

PART 1 of your worksheet

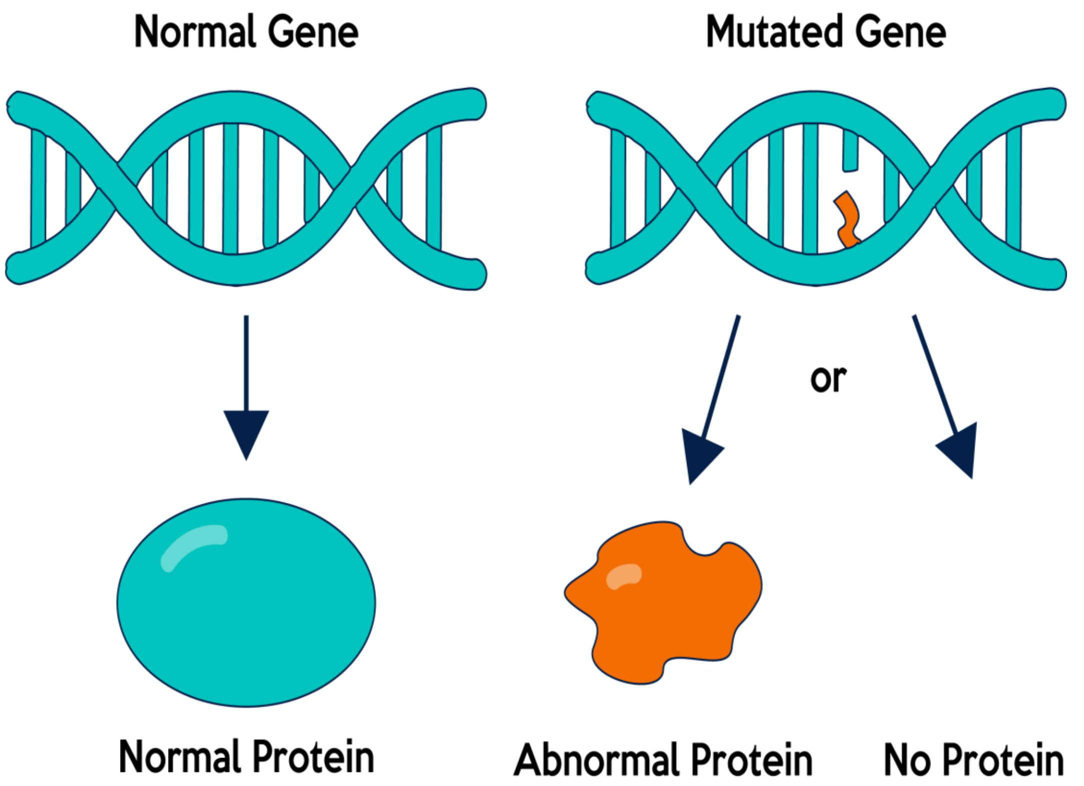
Behaviour can lead to mutagen exposure





...but limited evidence for effects in the germline (in humans).

Gouas et al. (2009)



normal phenotype

(mildly) abnormal phenotype

Unbuffered environment:
Mutation selected out

Buffered environment:
Mutation persists

Intro
Mutation
Gene flow
D+S

Does parental care cause mutations to accumulate?

wild-caught individuals

↓

experimental evolution

2 full care lines 2 no care lines

↓

20 generations

↓ ↓

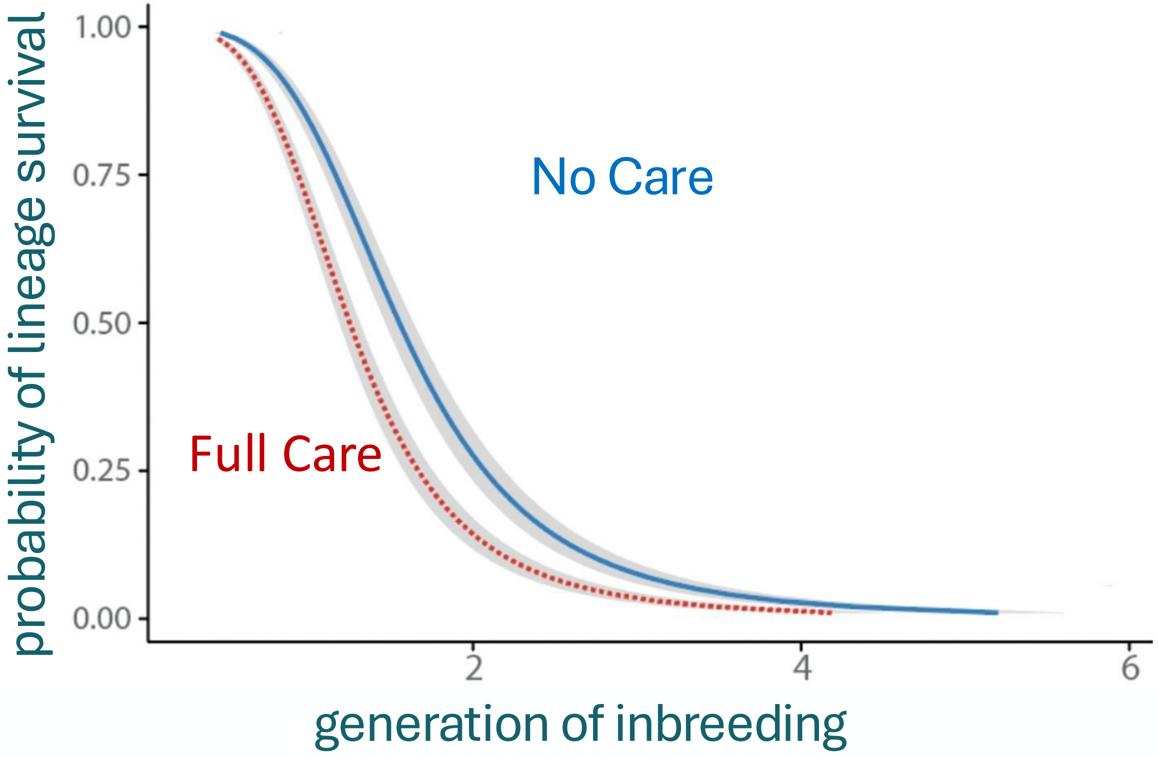
assessed deleterious mutations in each population by:

a) sequencing (Mashoodh et al. 2023)

b) inbreeding (Pascoal et al. 2023)

Inbreeding reveals higher mutation load in 'Full Cares'

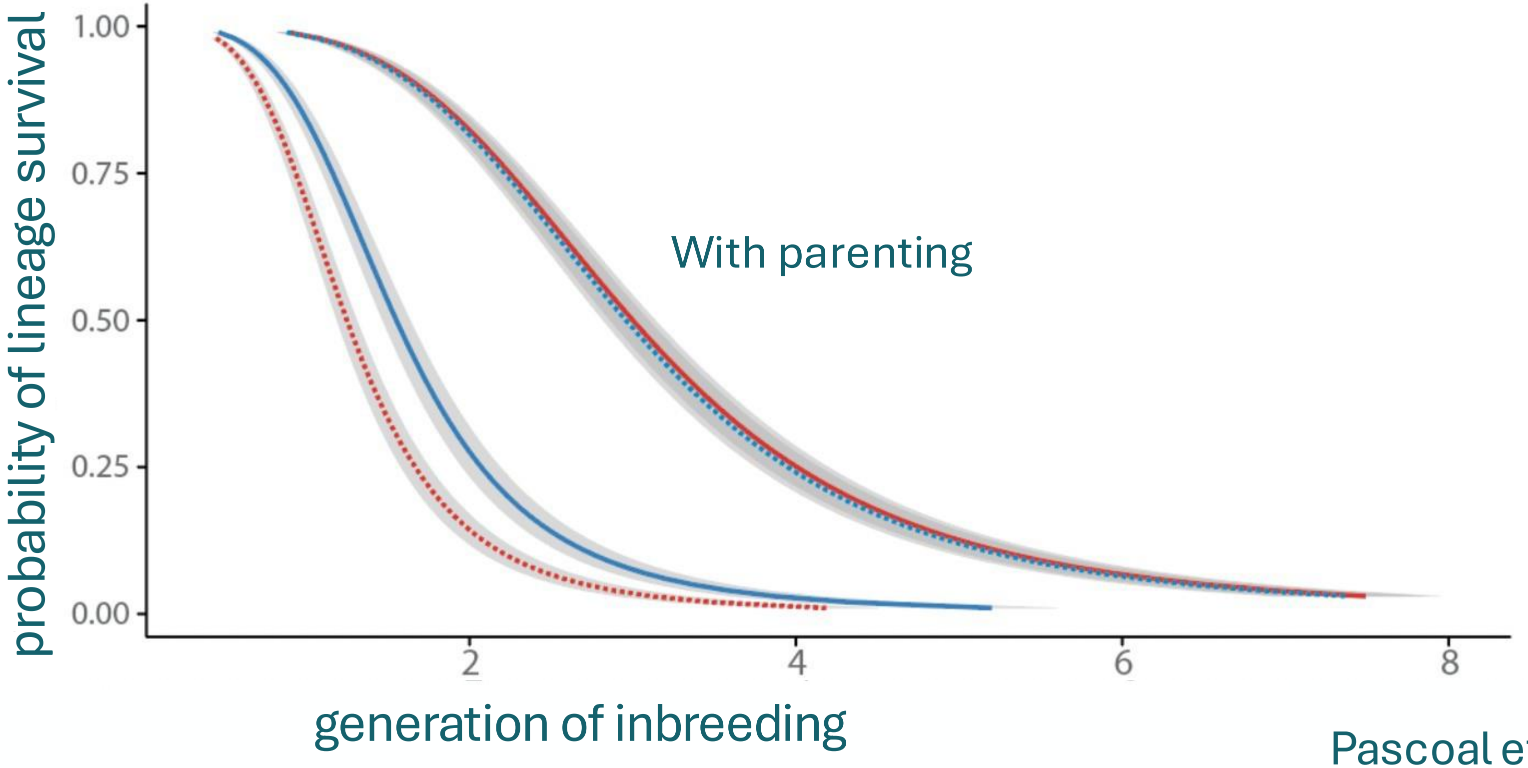
Survival of inbred lineage after 20 generations:



Pascoal et al. (2023)

Inbreeding reveals <u>higher mutation load</u> in 'Full Cares' ...but the cost of these mutations is offset by parenting!

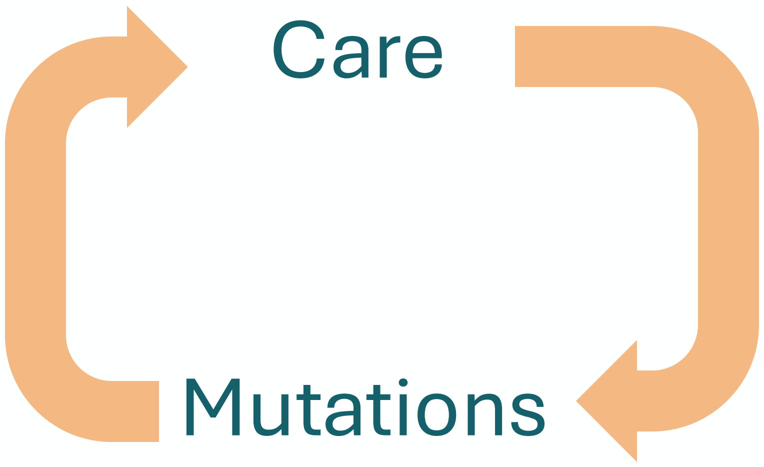
Survival of inbred lineage after 20 generations:



Pascoal et al. (2023)

Is care self-reinforcing?

- Care causes mildly deleterious mutations to accumulate…
- …but care is the phenotypic antidote for an increased mutation load



Pascoal et al. (2023)

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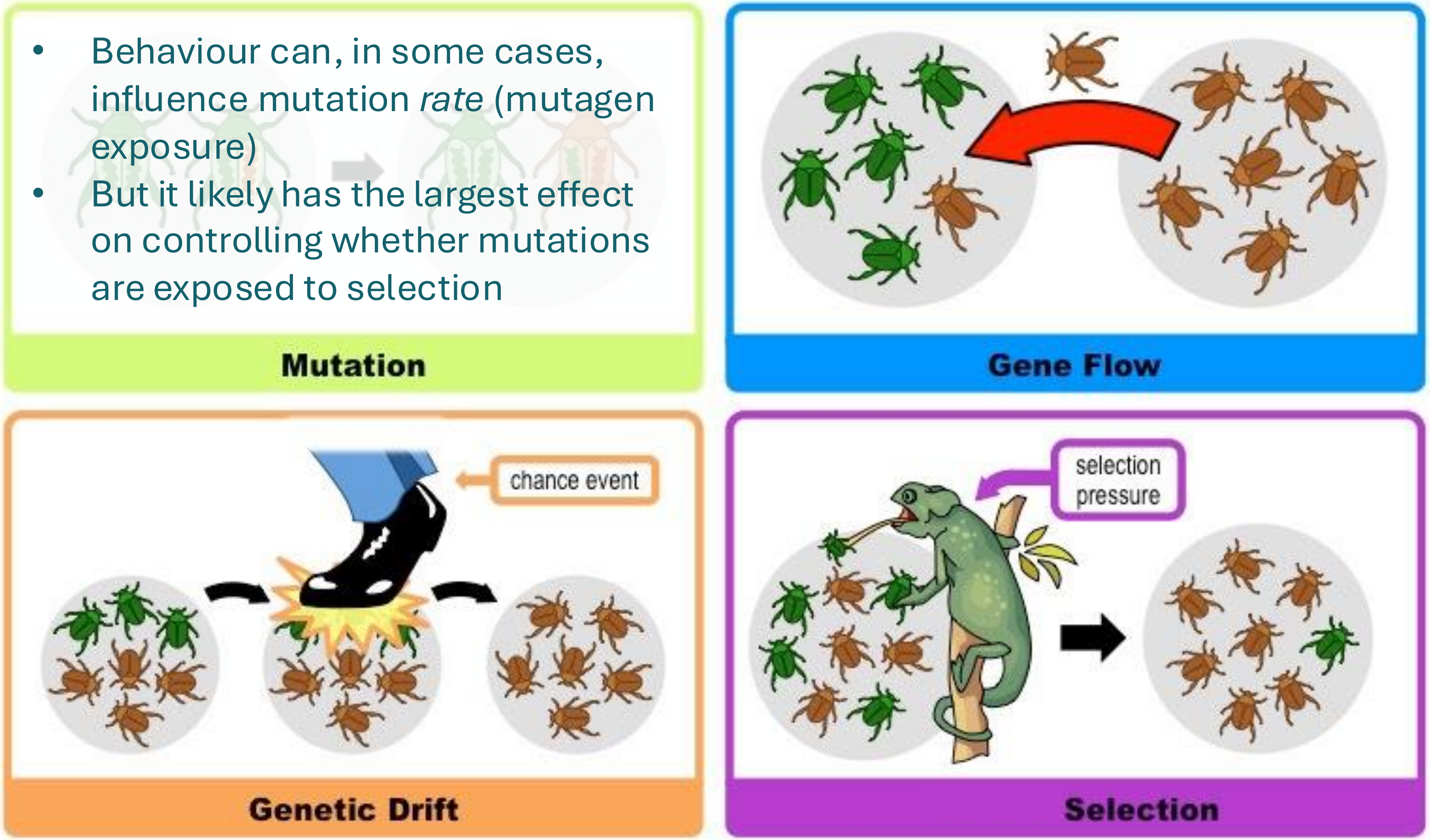
- But it likely has the largest effect on controlling whether mutations are exposed to selection



Intro
Mutation
Gene flow
D+S

Which **cultural** processes might influence drift, gene flow, and/or selection?

- Marriage practices
- Inheritance of power and property
- Language and identity
- Warfare
- Beauty standards

Matri/Patrilocality: Male vs female migration network

Endogamy/exogamy: isolation and local drift (e.g. Indian castes) vs more mixing

Group identity: Rate of admixture on contact between groups.

Polygyny/polyandry: reproductive skews (more skew -> more drift)

Power skews: Culturally inherited reproductive success (drift)

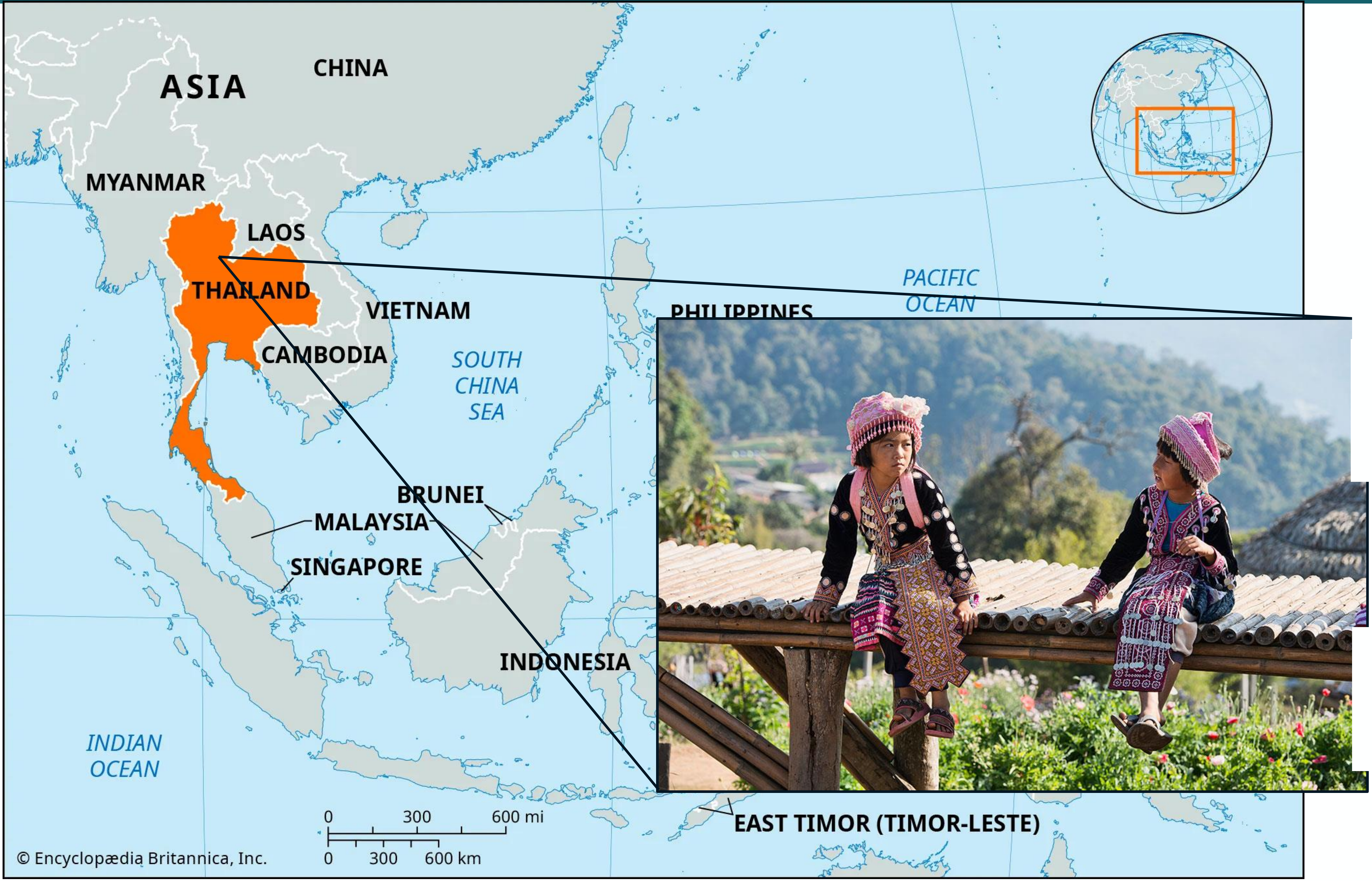
Dispersal tendencies: When are villages/bands founded/go extinct? When do groups/individuals migrate?

Population densities/sizes: families vs bands vs villages vs cities

Assortative/disassortative mating

Marriage traditions leave distinct signals in the genome... but not all of the genome!

Marriage practices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MYANMAR
LAOS
THAILAND
VIETNAM
CAMBODIA
PHILIPPINES
PACIFIC OCEAN
SOUTH CHINA SEA
BRUNEI
MALAYSIA
SINGAPORE
INDONESIA
INDIAN OCEAN
EAST TIMOR (TIMOR-LESTE)
0 300 600 mi
0 300 600 km
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Karen

♀ ♂
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♀ ♂
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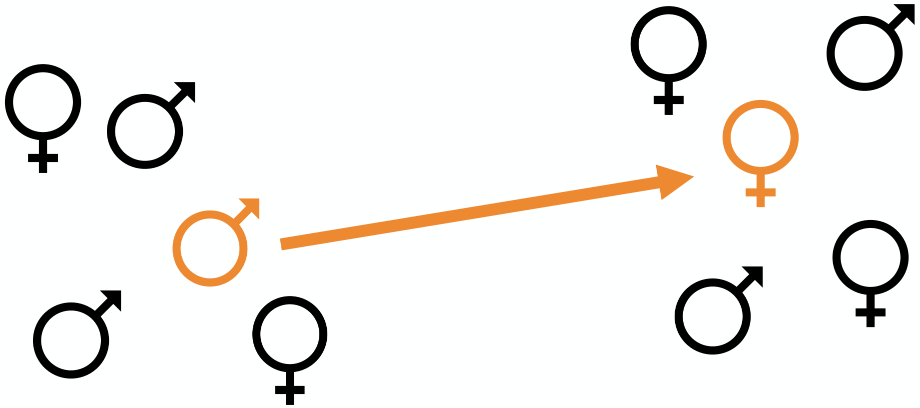
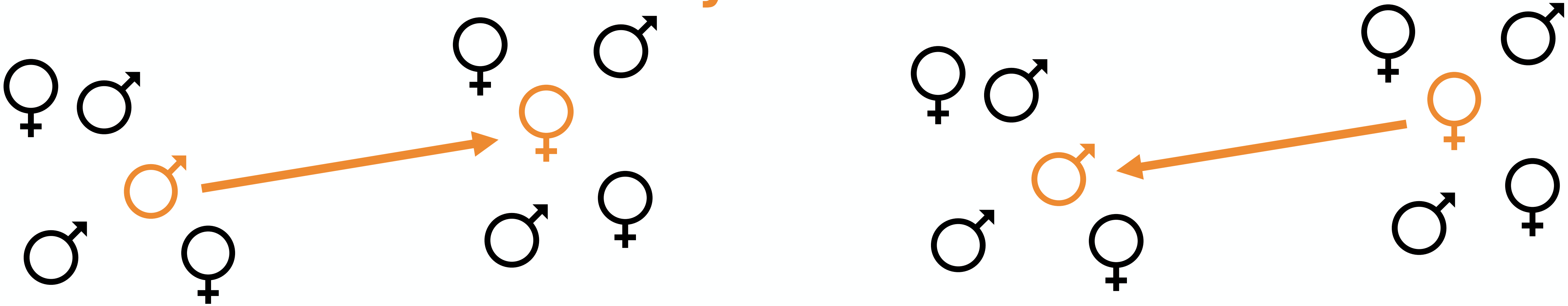
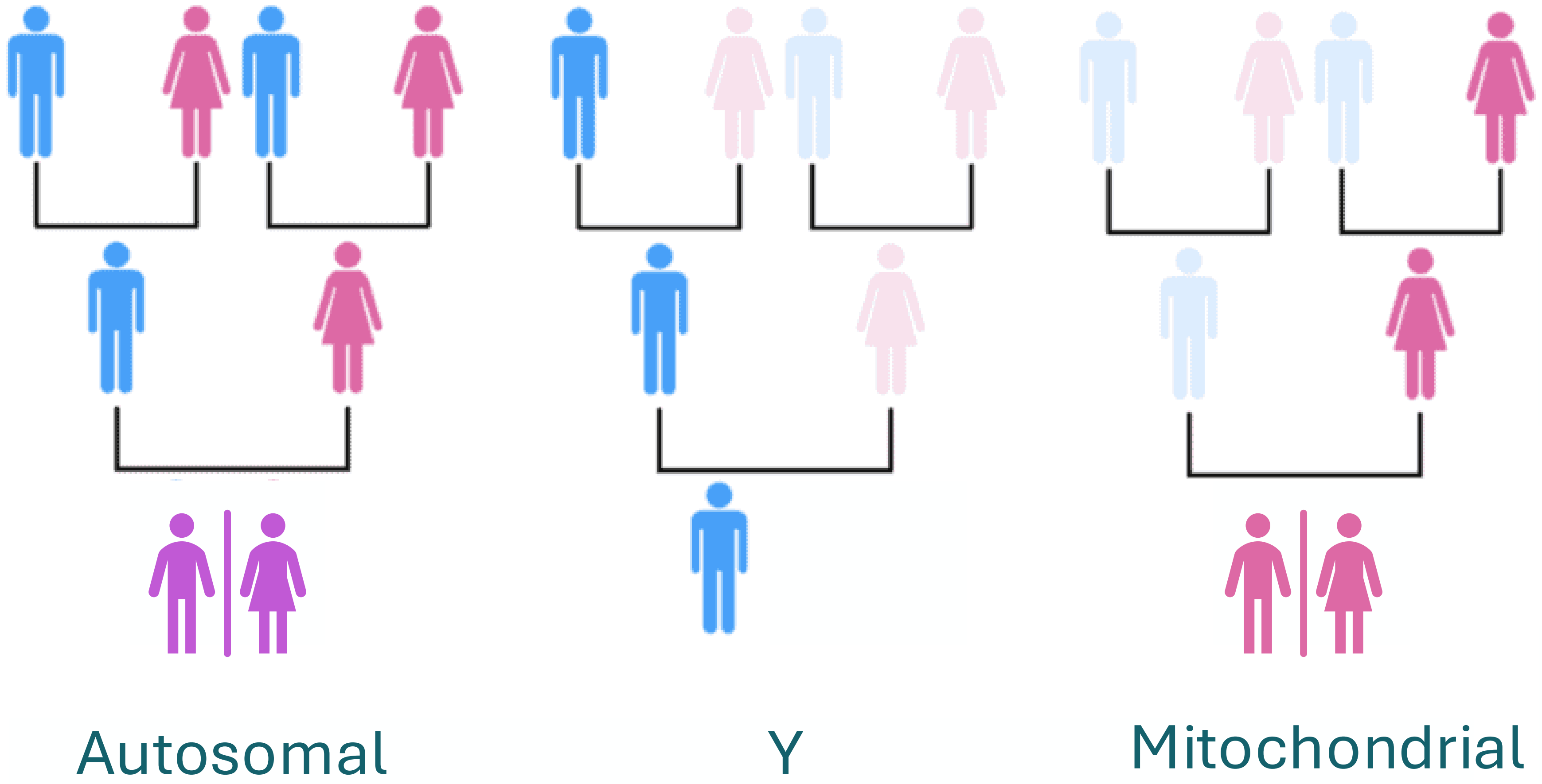
Karen



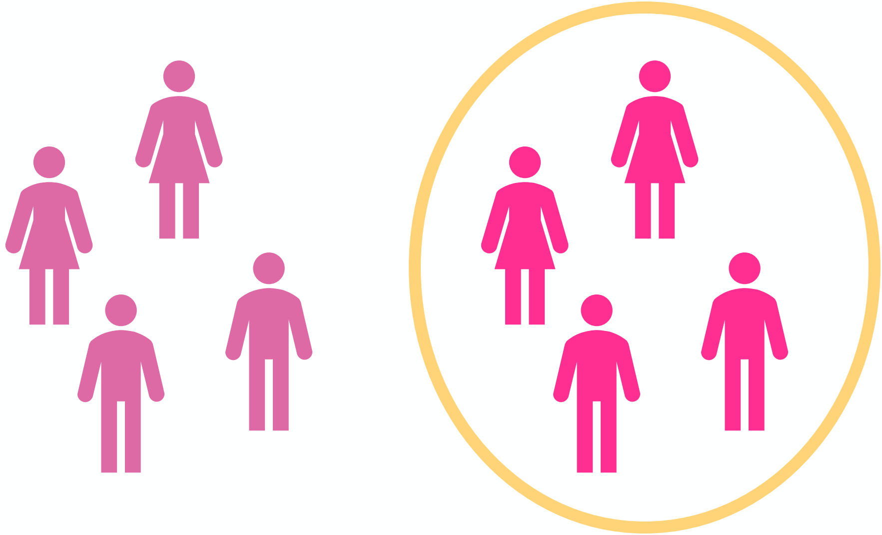
PART 2 of your worksheet

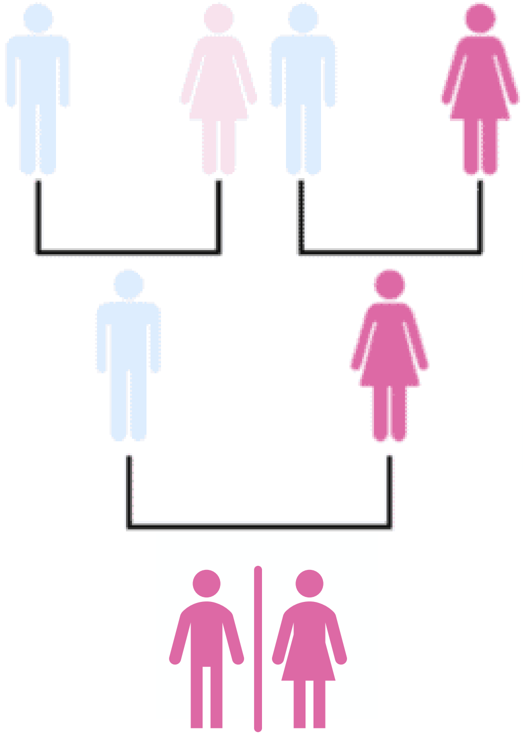



Autosomal
Y
Mitochondrial

Patrilocal

Mitochondrial: 1

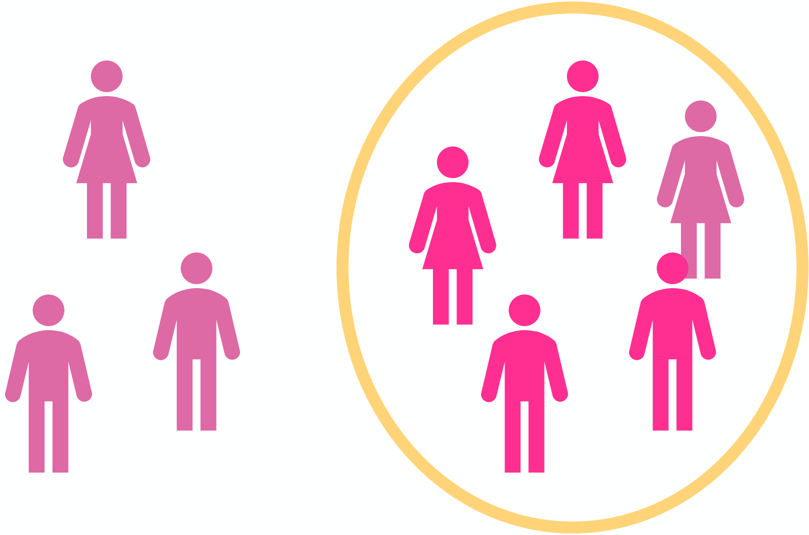


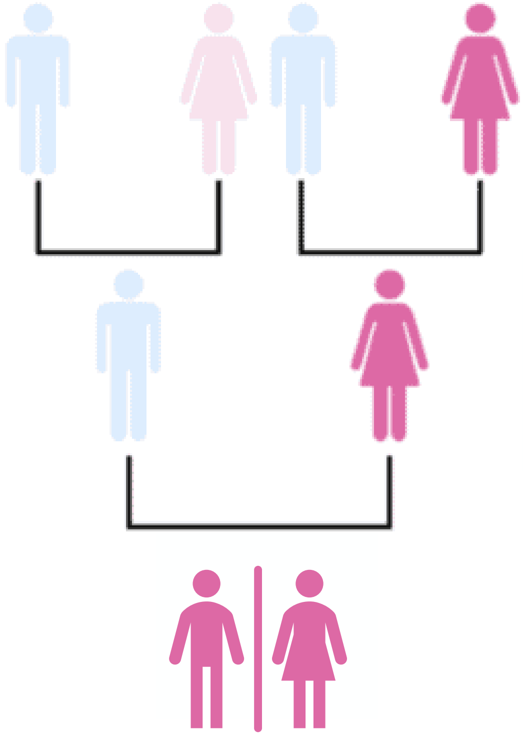


Mitochondrial

Patrilocal

Mitochondrial: 2

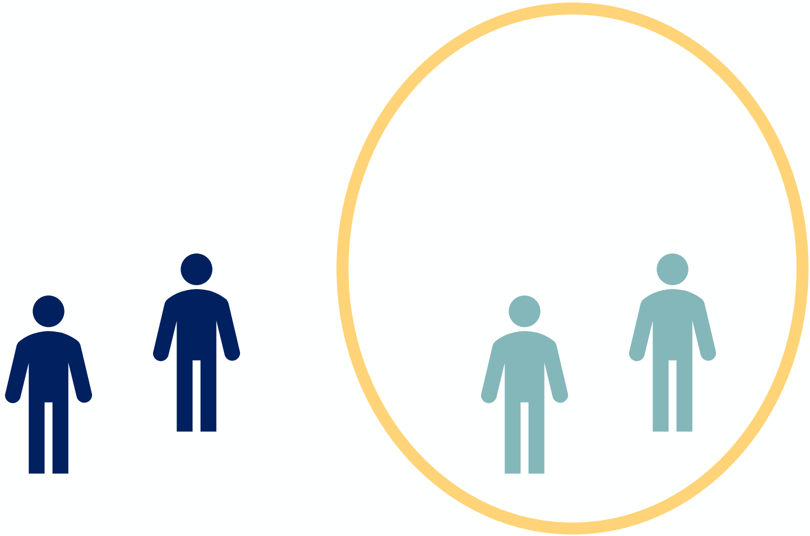


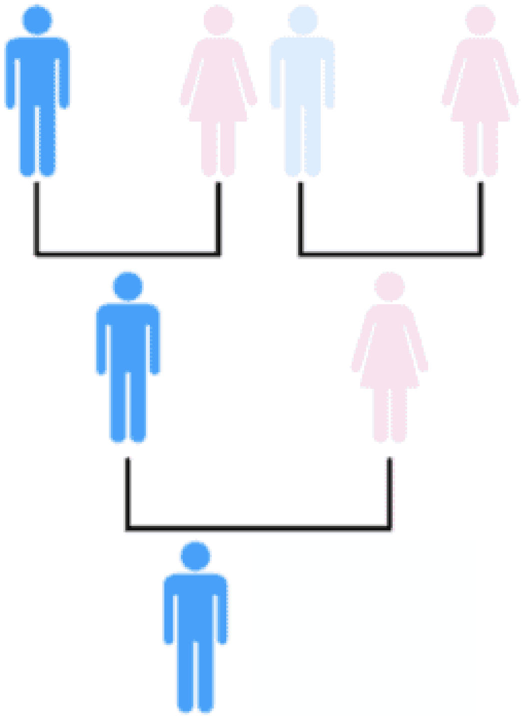


Mitochondrial

Patrilocal

Mitochondrial: 2
Y-chromosome: 1

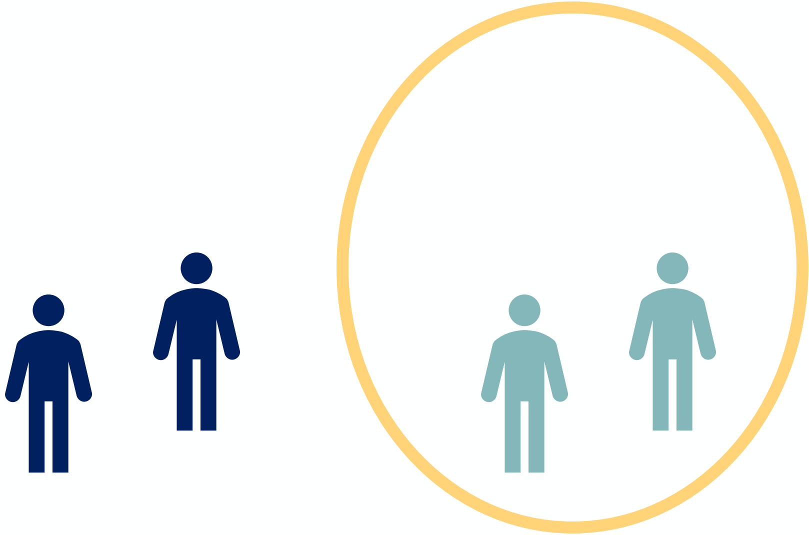


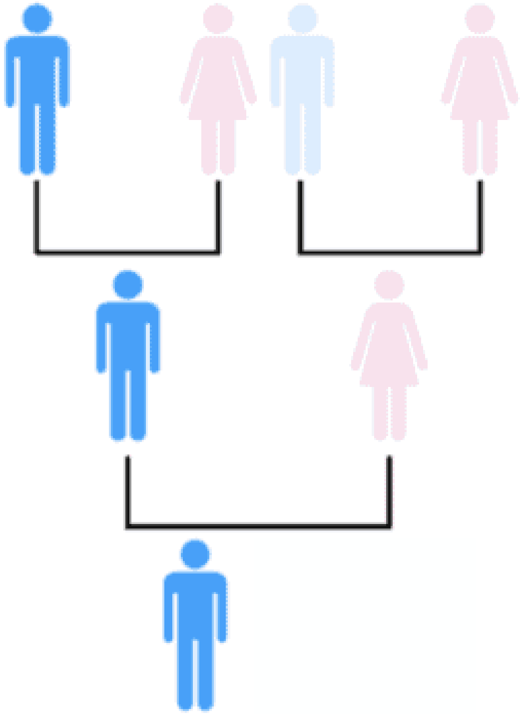


Y

Patrilocal: expect lower Y-chromosome diversity

Mitochondrial: 2
Y-chromosome: 1

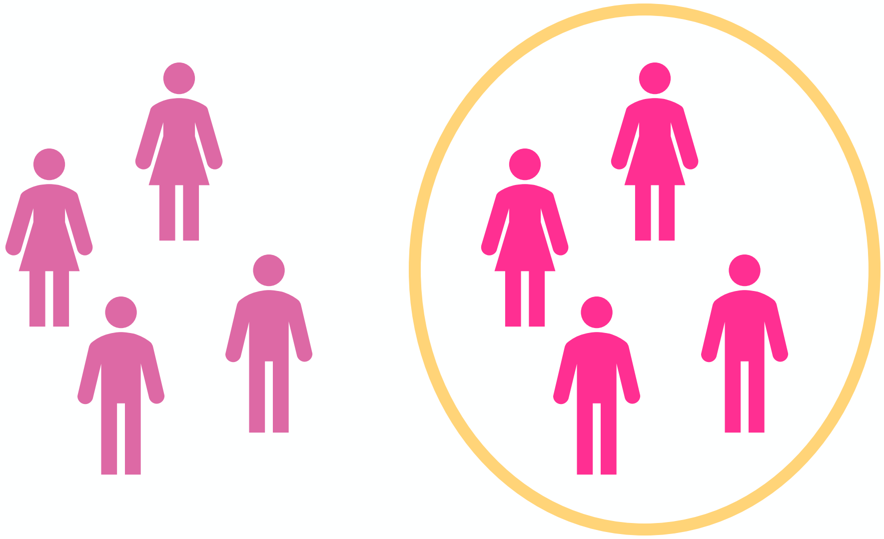


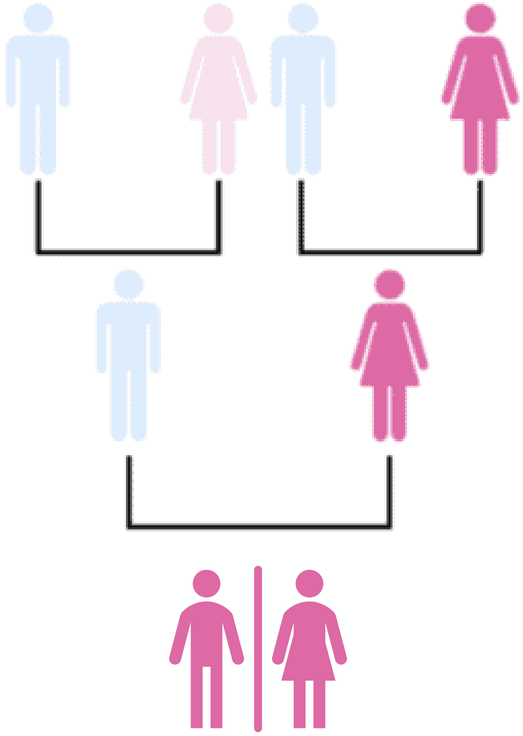


Y

Matrilocal

Mitochondrial: 1

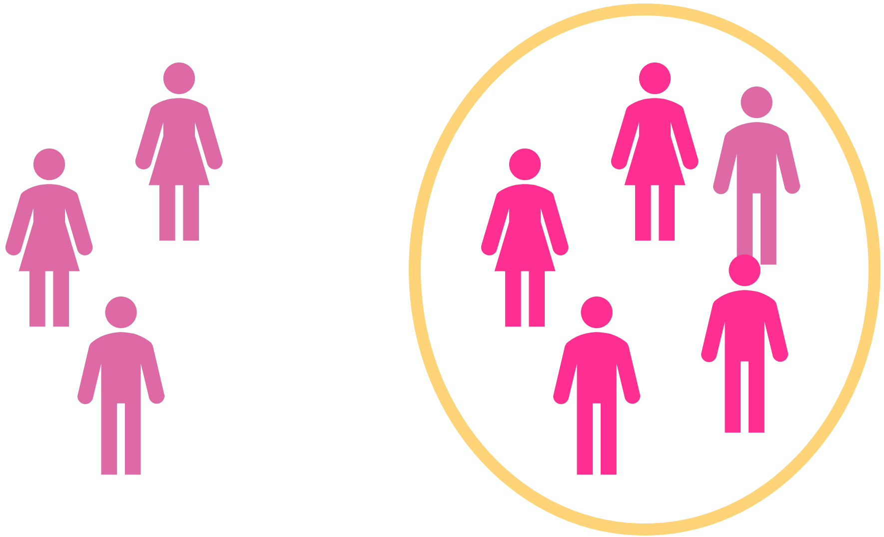


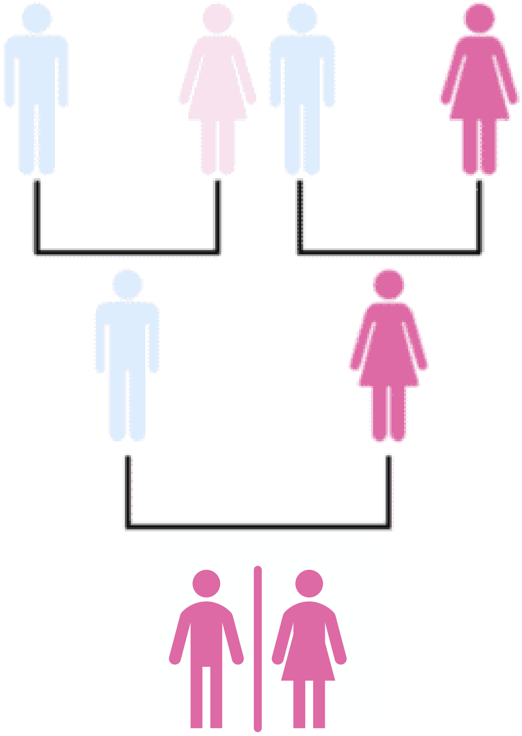


Mitochondrial

Matrilocal

Mitochondrial: 2

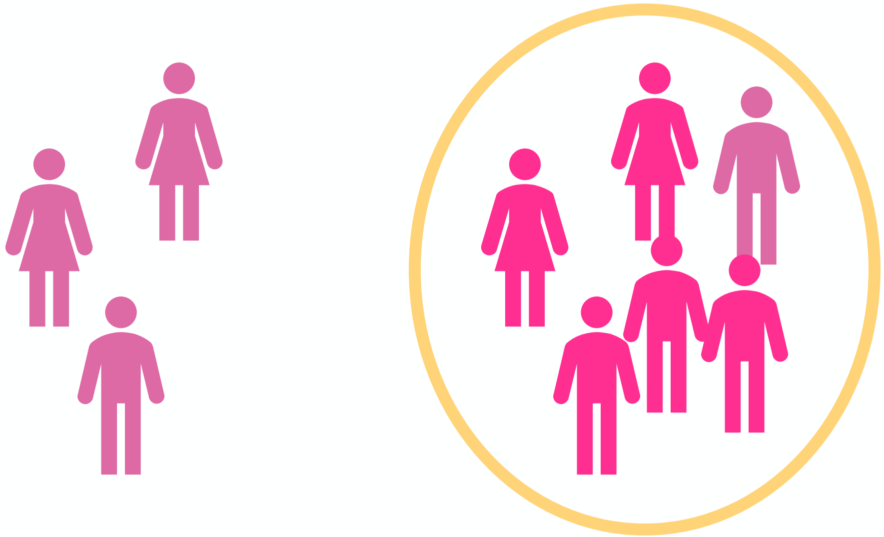


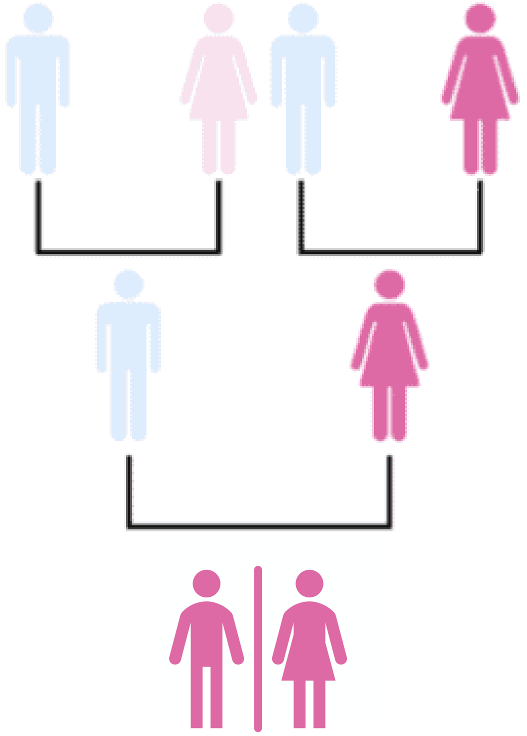


Mitochondrial

Matrilocal

Mitochondrial: 2

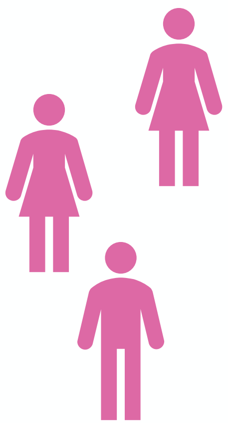


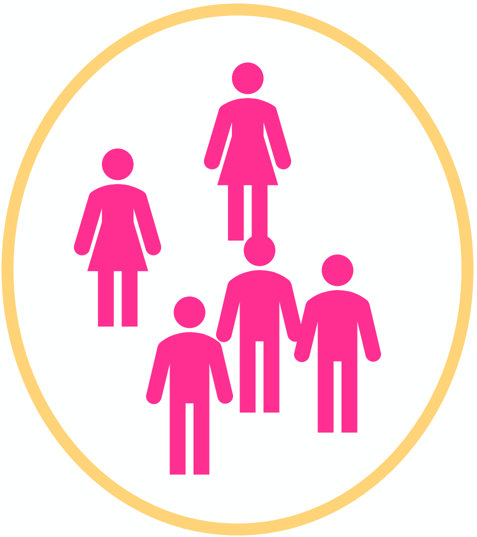


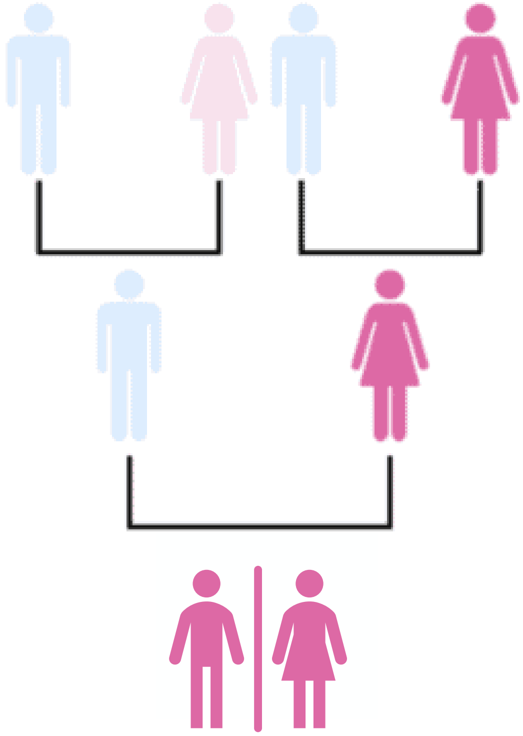
Mitochondrial

Matrilocal

Mitochondrial: 1 (long run)



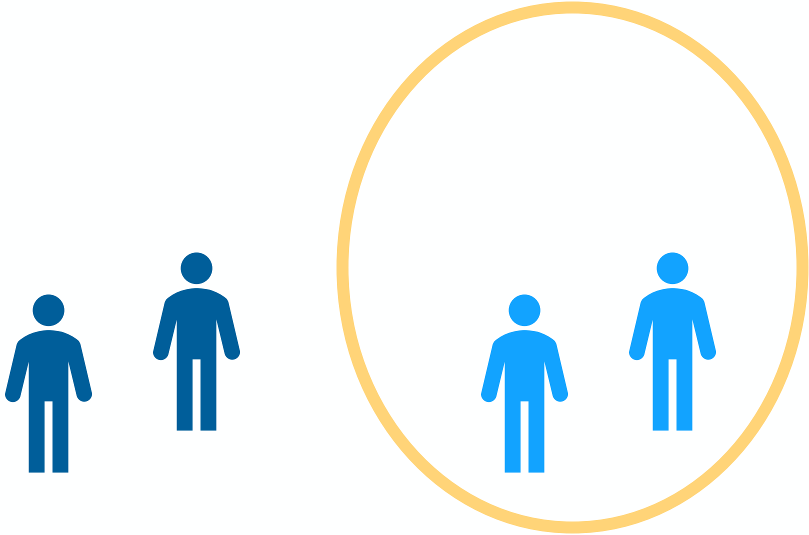


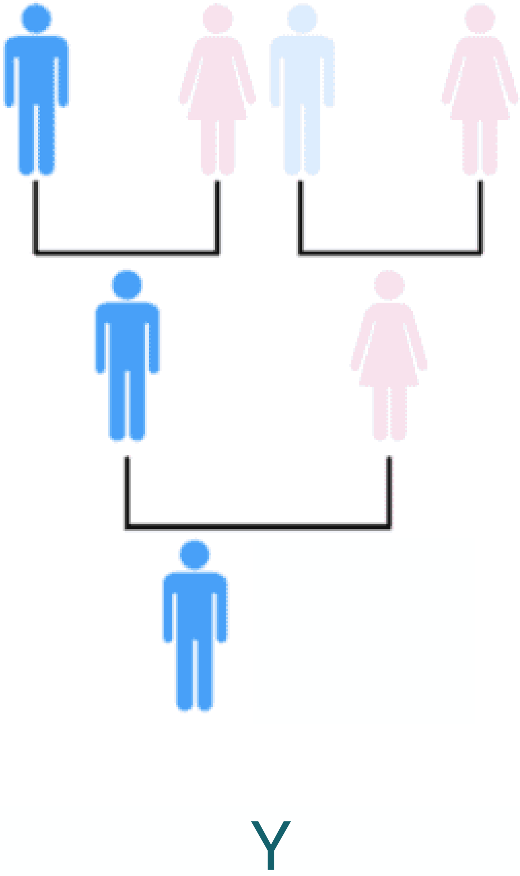


Mitochondrial

Matrilocal

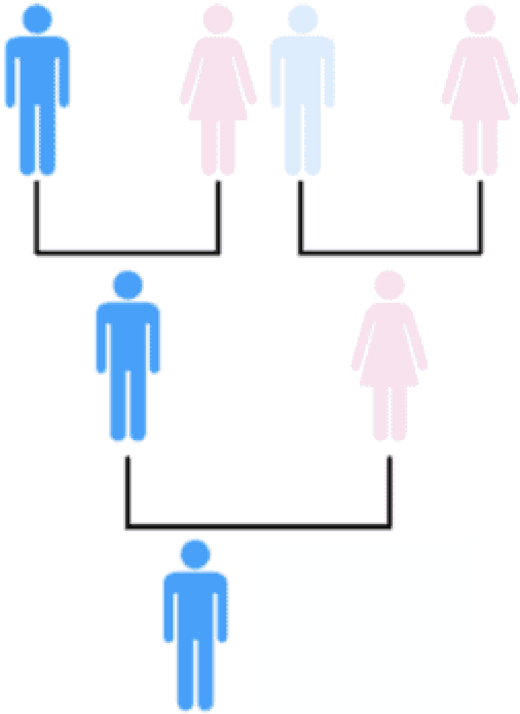
Mitochondrial: 1 (long run)
Y-chromosome: 1





Matrilocal

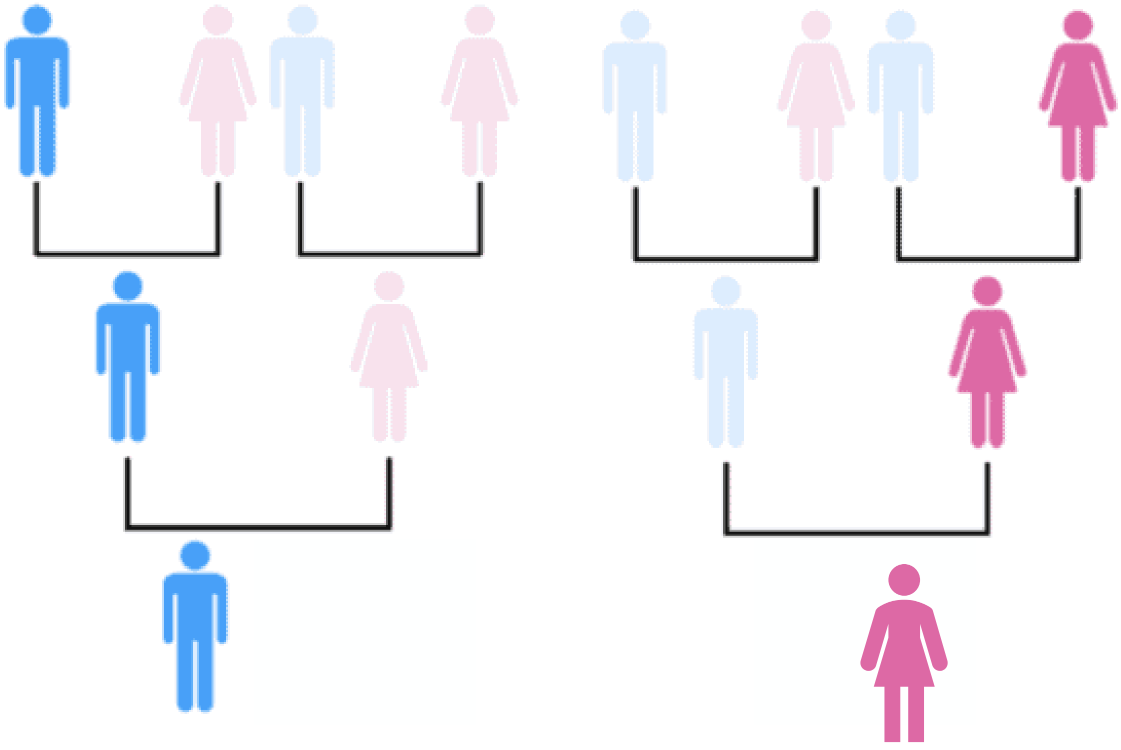
Mitochondrial: 1 (long run)
Y-chromosome: 2



Y

Matrilocal: expect greater Y-chromosome diversity

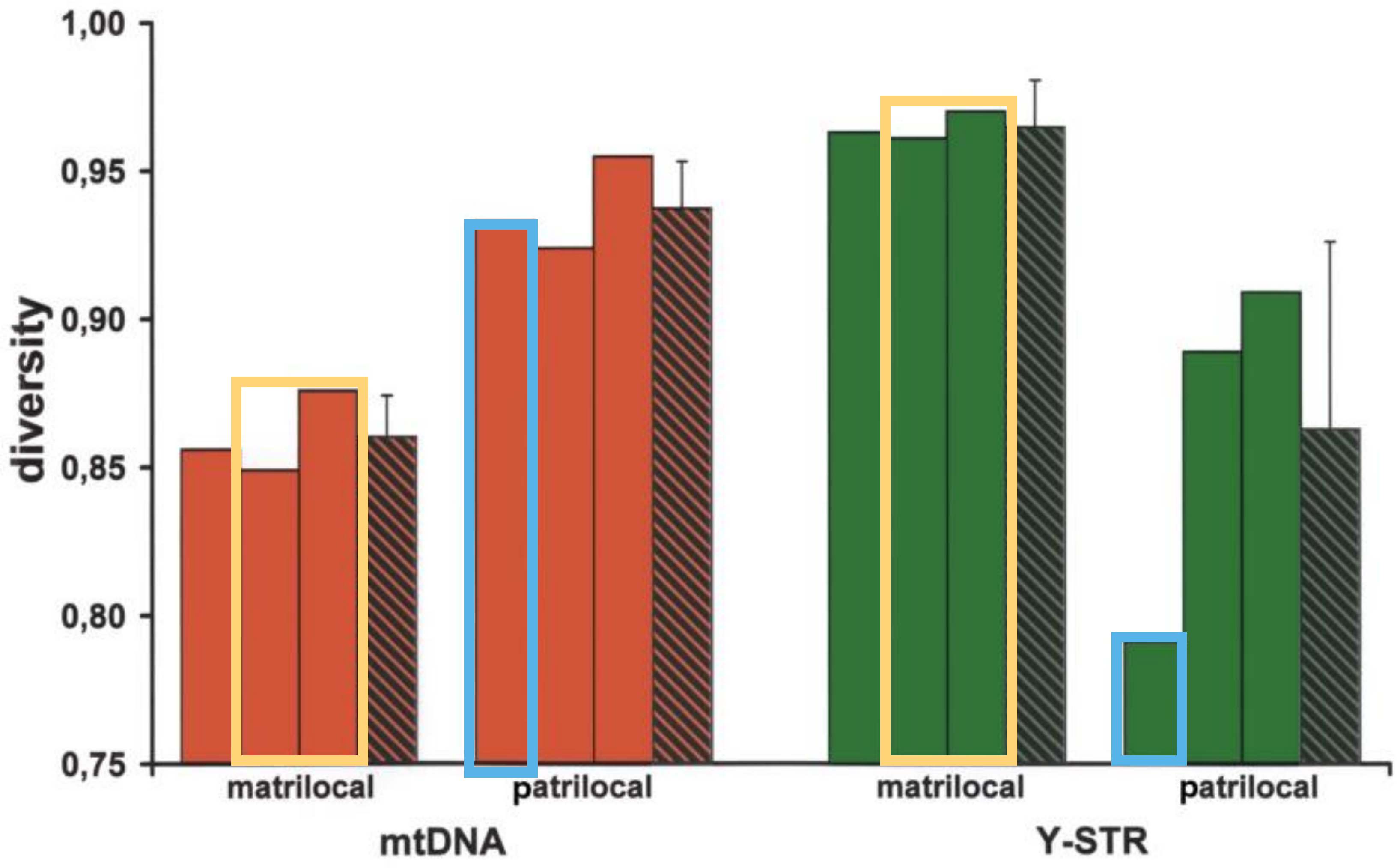
Mitochondrial: 1 (long run)
Y-chromosome: 2 (long run)



Y

Mitochondrial

Diversity reflects transmissible copies

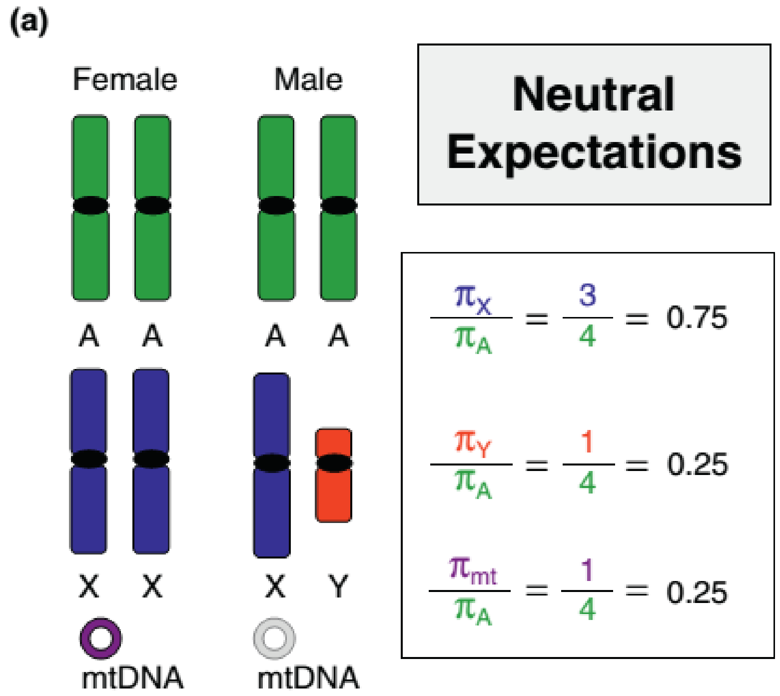


Matrilocal: expect greate Y-chromosome diversity
- mtDNA less diverse
- Y more diverse

Patrilocal: expect greater mtDNA diversity
- mtDNA more diverse
- Y less diverse

What is going on?

Key



Under random mating and neutrality, diversity of autosomes/ X/Y/mtDNA follow the **ploidy of transmissible copies** in the population

Neutral expectations:

- **N_e or diversity follows a 4:3:1:1 ratio across the autosomes, X, Y, and mtDNA respectively**
  - That is, autosomal diversity is expected to be 4x higher than mtDNA diversity; and Y and mtDNA are expected to be equally diverse

Sex-biased processes drive divergence from these expectations.

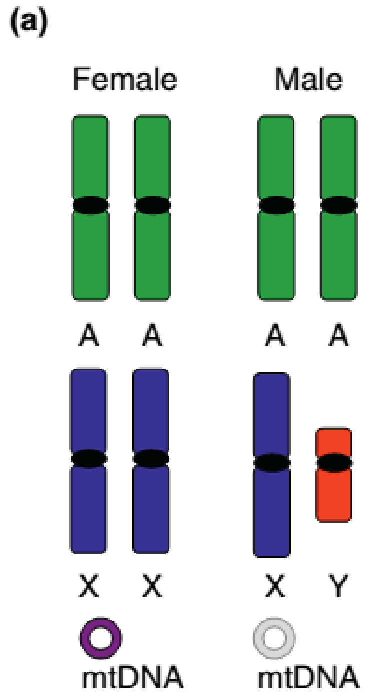
- Thus, comparing diversity of loci can provide a view into male vs female demography.

Webster and Wilson Sayres 2016 Curr Opin Genet Dev

What is going on?

Key

(a)



Neutral Expectations

$$\frac{\pi_X}{\pi_A} = \frac{3}{4} = 0.75$$

$$\frac{\pi_Y}{\pi_A} = \frac{1}{4} = 0.25$$

$$\frac{\pi_{mt}}{\pi_A} = \frac{1}{4} = 0.25$$

Under random mating and neutrality, diversity of autosomes/X/Y/mtDNA follow the **ploidy of transmissible copies** in the population

Neutral expectations:

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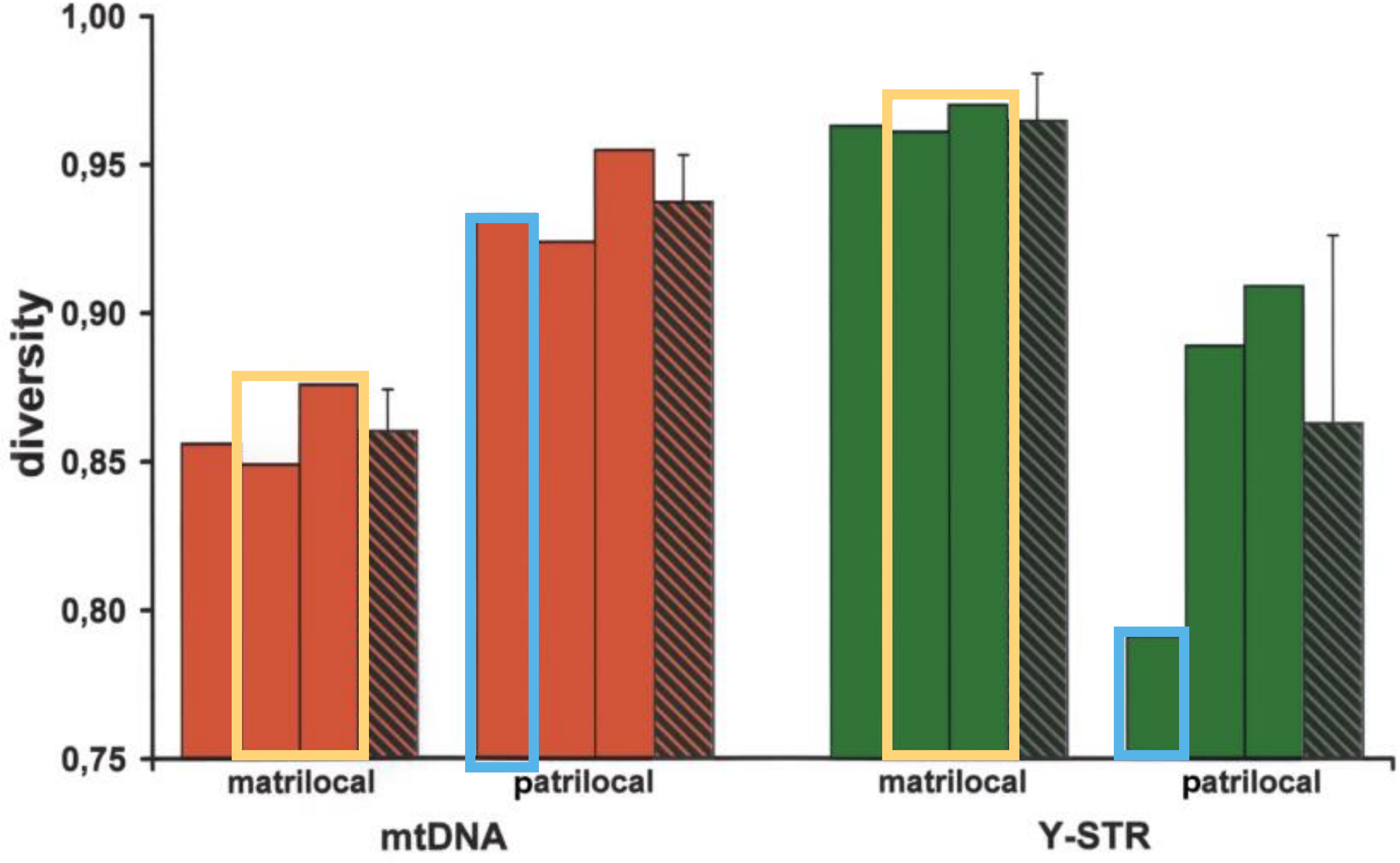
Sex-biased processes drive divergence from these expectations.

Reproductive skew

Marriage

- Unequal sex ratios (or large differences in effective population sizes between the sexes)
- Sex-biased migration and dispersal patterns
- Sex differences in generation time
- Sex-biased admixture and introgression

Webster and Wilson Sayres 2016 Curr Opin Genet Dev



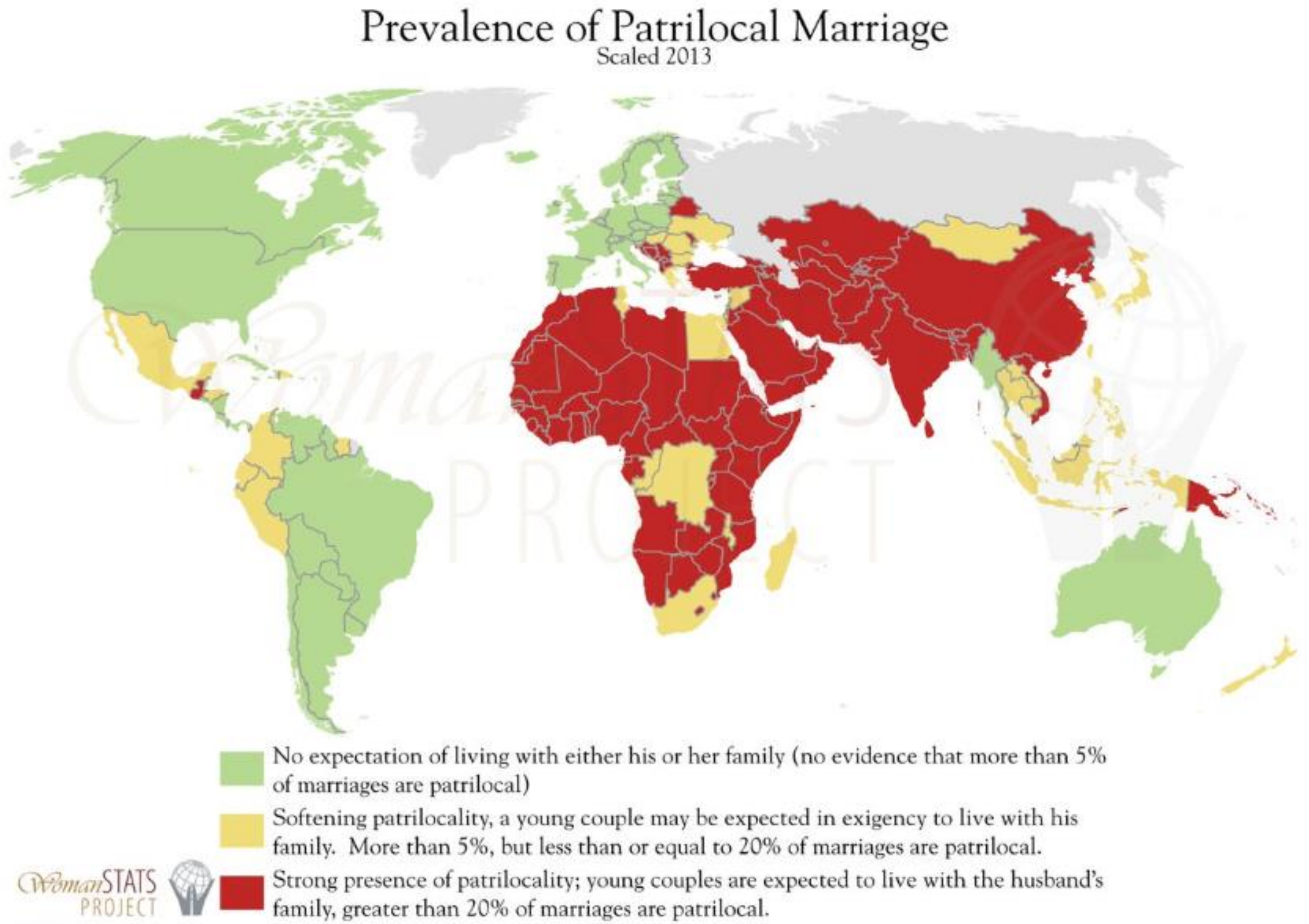
Matrilocal: expect greate Y-chromosome diversity

- mtDNA less diverse
- Y more diverse

Patrilocal: expect greater mtDNA diversity

- mtDNA more diverse
- Y less diverse

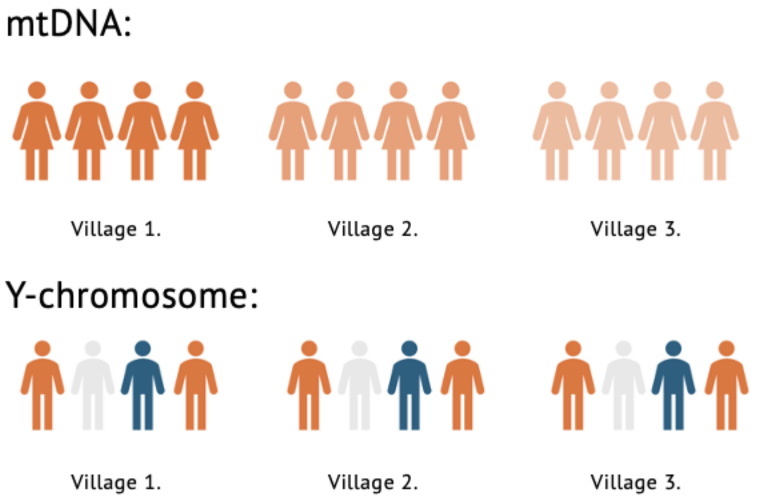
Do the same patterns hold at a global scale?

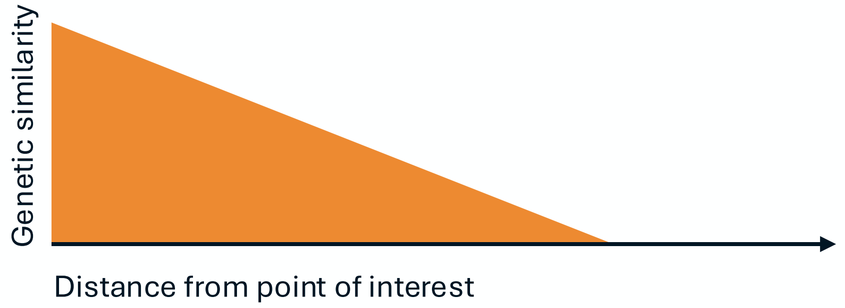


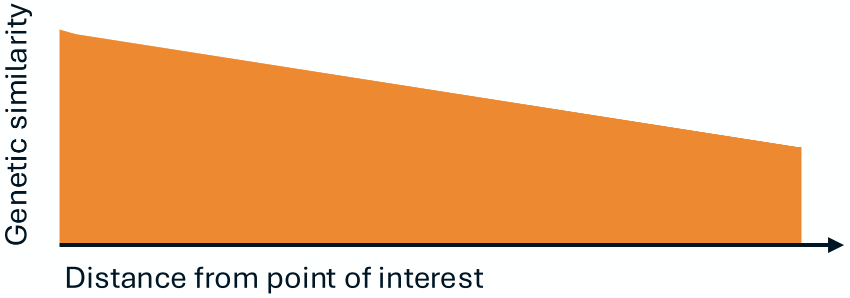
Matrilocal: expect greater Y-chromosome diversity *at small scales!*

Patrilocal: expect greater mtDNA diversity *at small scales!*

Do the same patterns hold at a global scale?





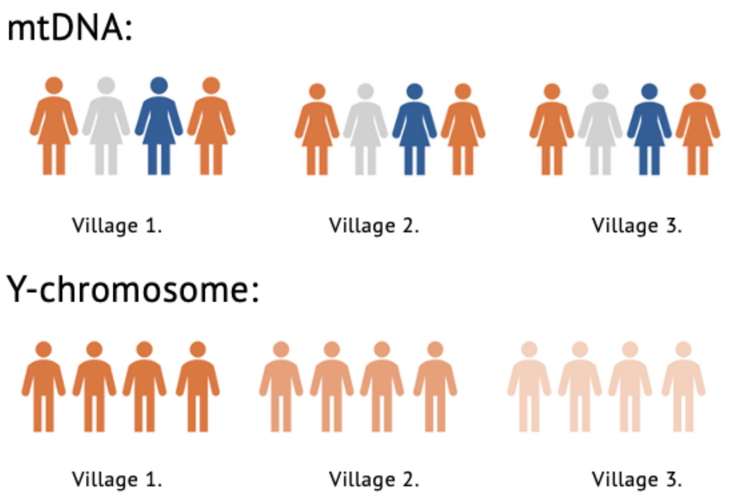


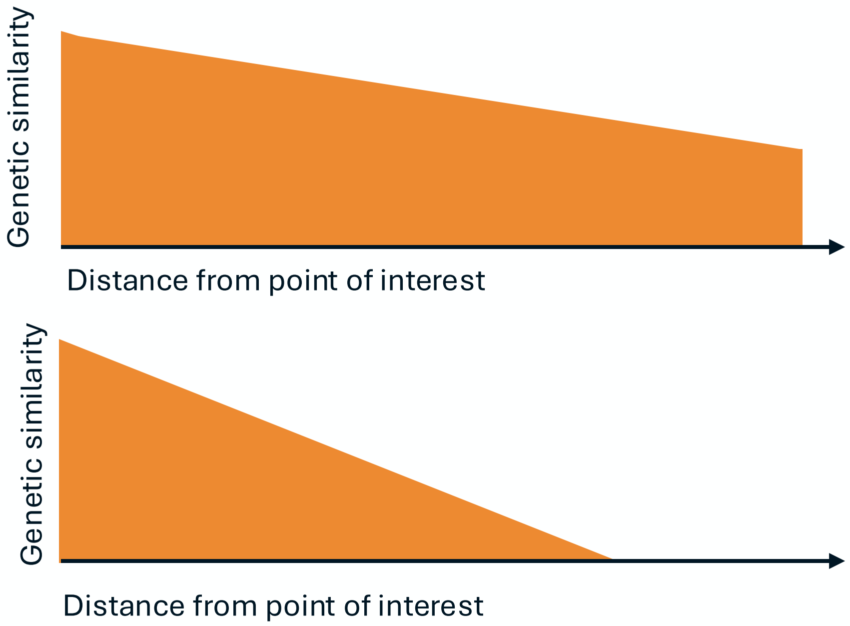
Matrilocal: expect greater Y-chromosome diversity
at small scales!

Patrilocal: expect greater mtDNA diversity
at small scales!

Do the same patterns hold at a global scale?

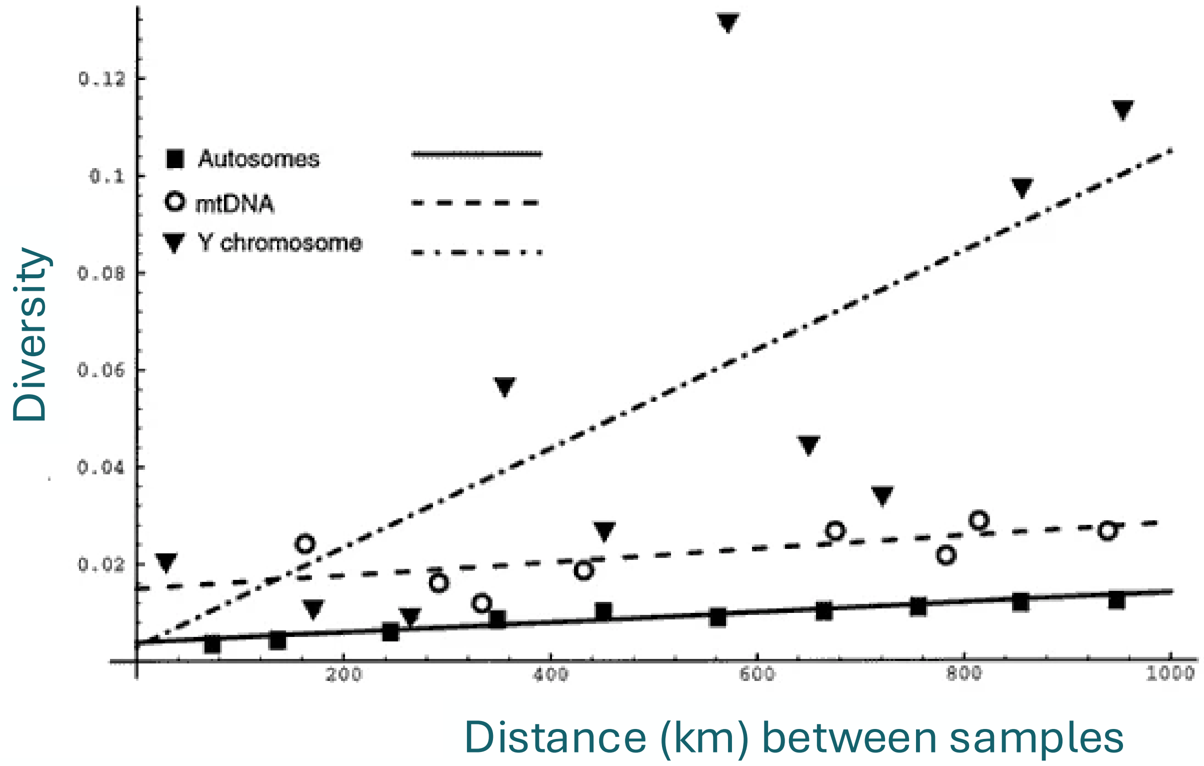
Matrilocal: expect greater Y-chromosome diversity
at small scales!

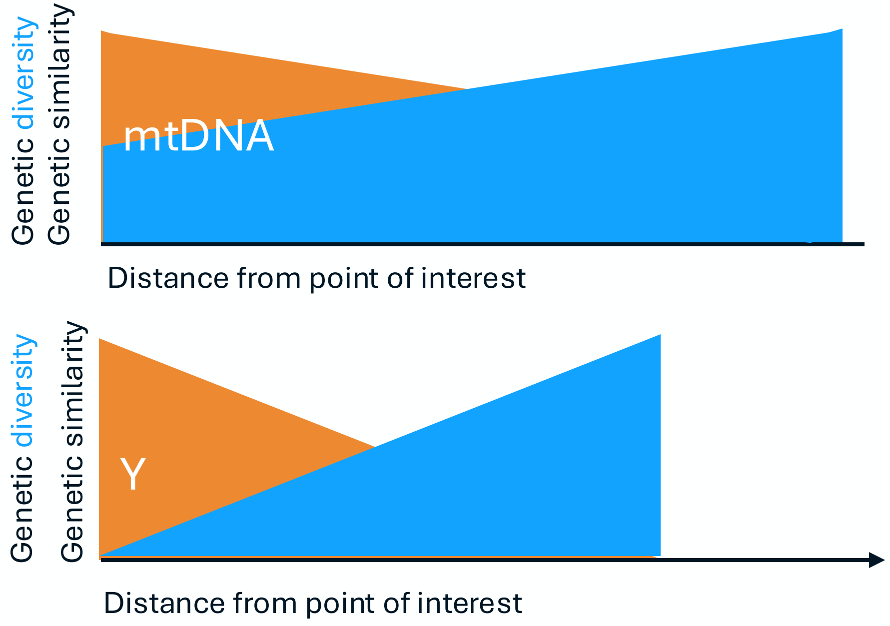




Patrilocal: expect greater mtDNA diversity
at small scales!

Do the same patterns hold at a global scale?





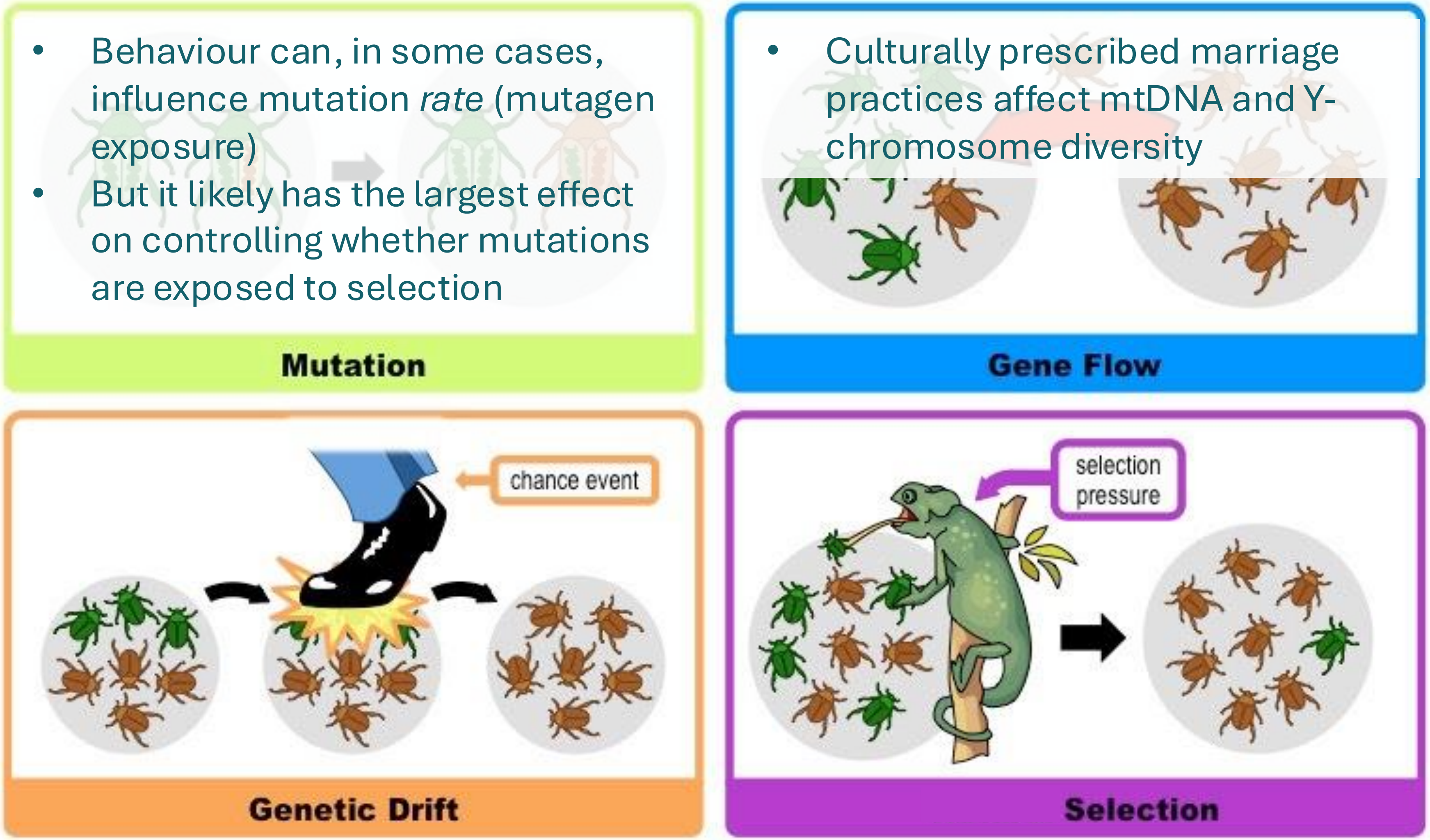
Low mtDNA and high Y-chromosome differences *between groups*, concordant with patrilocality

Seielstad et al. (1998)

Which of these processes might be influenced by **culture** (or behaviour more generally)?

- Behaviour can, in some cases, influence mutation *rate* (mutagen exposure)
- But it likely has the largest effect on controlling whether mutations are exposed to selection

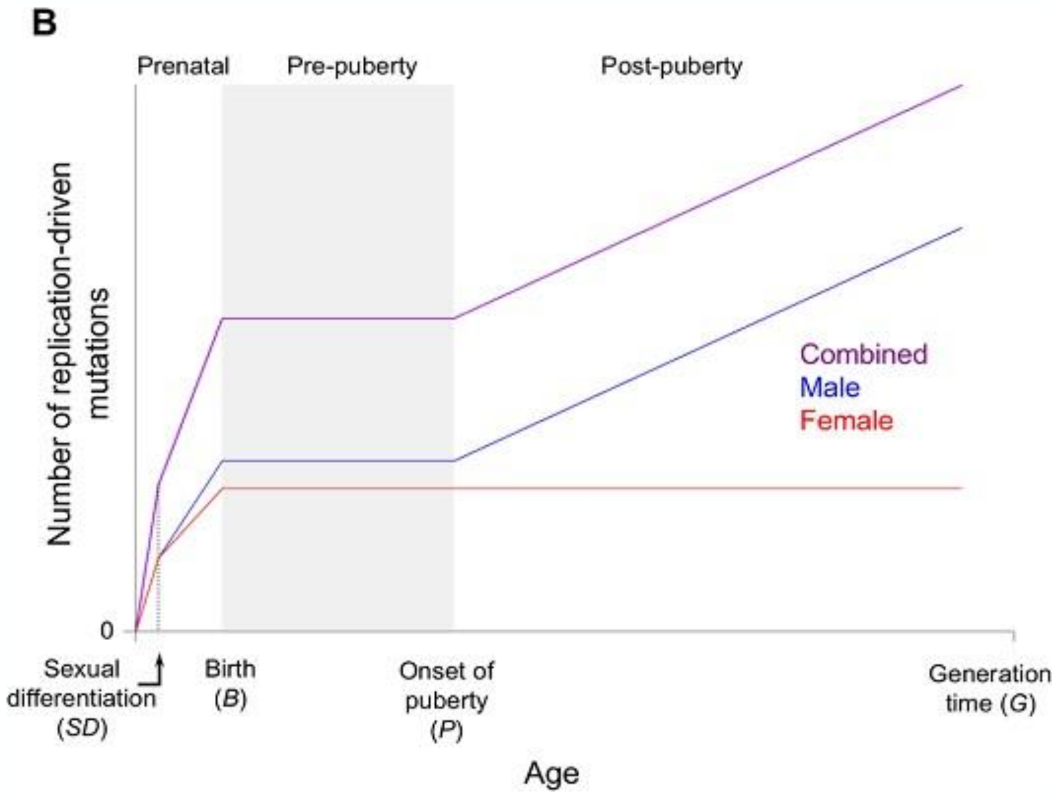
- Culturally prescribed marriage practices affect mtDNA and Y-chromosome diversity

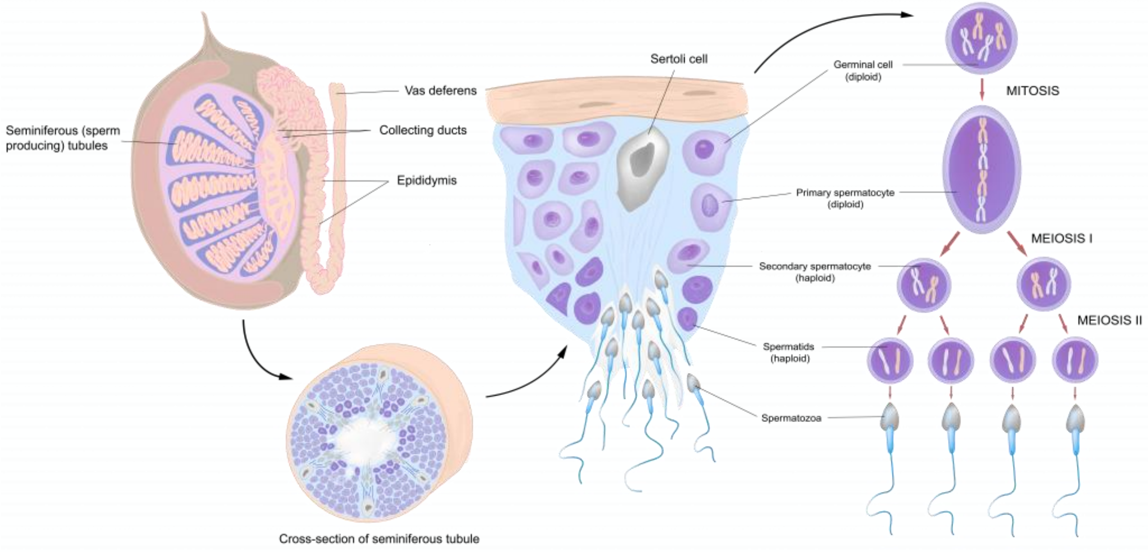


Intro
Mutation
Gene flow
D+S

Not just culture!

Biological drivers of different diversity patterns on mtDNA, X- and Y-chromosomes



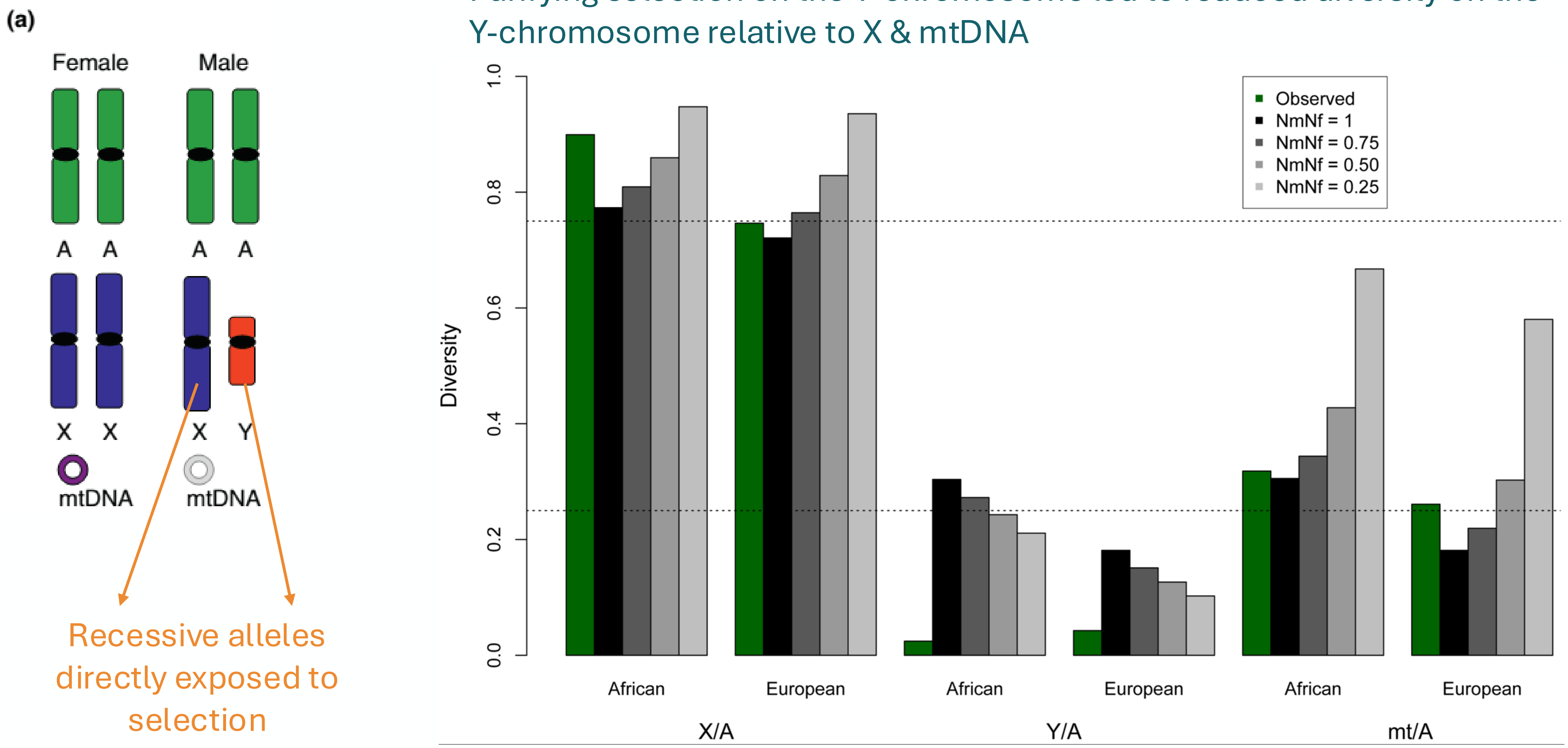


Male mutation bias

- Greater number of germline cell divisions in males than in females
- Sperm produced throughout life, and requires mitosis
- Increases diversity in this order: Y chromosome >> autosomes > X chromosome

Biological drivers of different diversity patterns on mtDNA, X- and Y-chromosomes

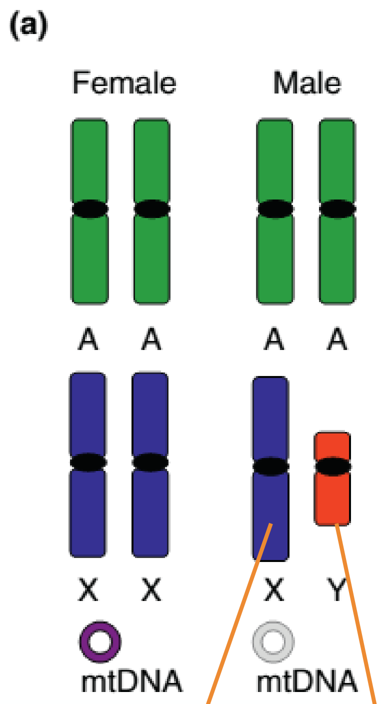
Purifying selection on the Y-chromosome led to reduced diversity on the Y-chromosome relative to X & mtDNA



Wilson Sayres et al. (2014)

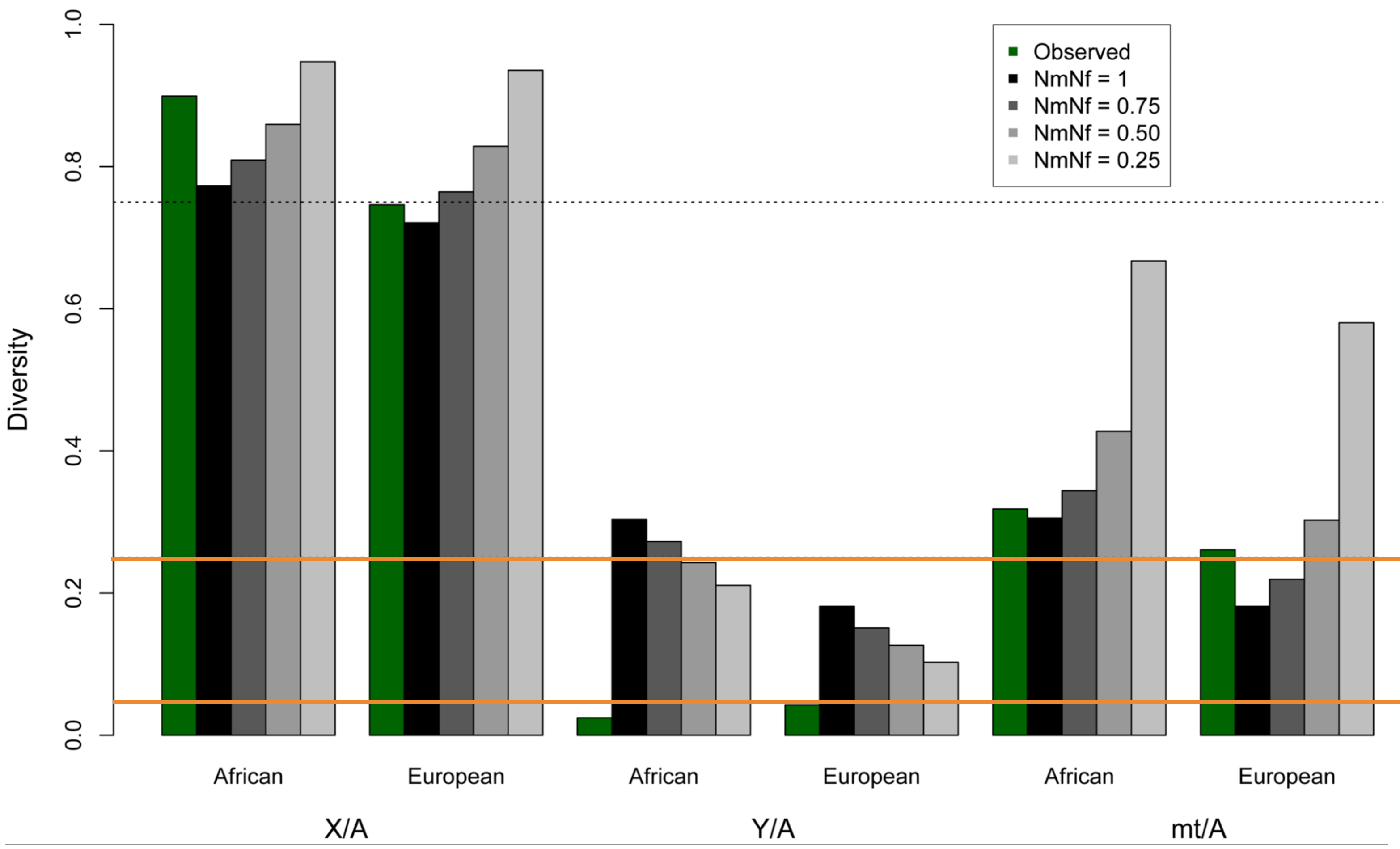
Not just culture!

Biological drivers of different diversity patterns on mtDNA, X- and Y-chromosomes



Recessive alleles directly exposed to selection

Purifying selection on the Y-chromosome led to reduced diversity on the Y-chromosome relative to X & mtDNA



Expected for Y

Observed for Y

Wilson Sayres et al. (2014)

Which of these processes might be influenced by **culture** (or behaviour more generally)?

- Behaviour can, in some cases, influence mutation *rate* (mutagen exposure)
- But it likely has the largest effect on controlling whether mutations are exposed to selection

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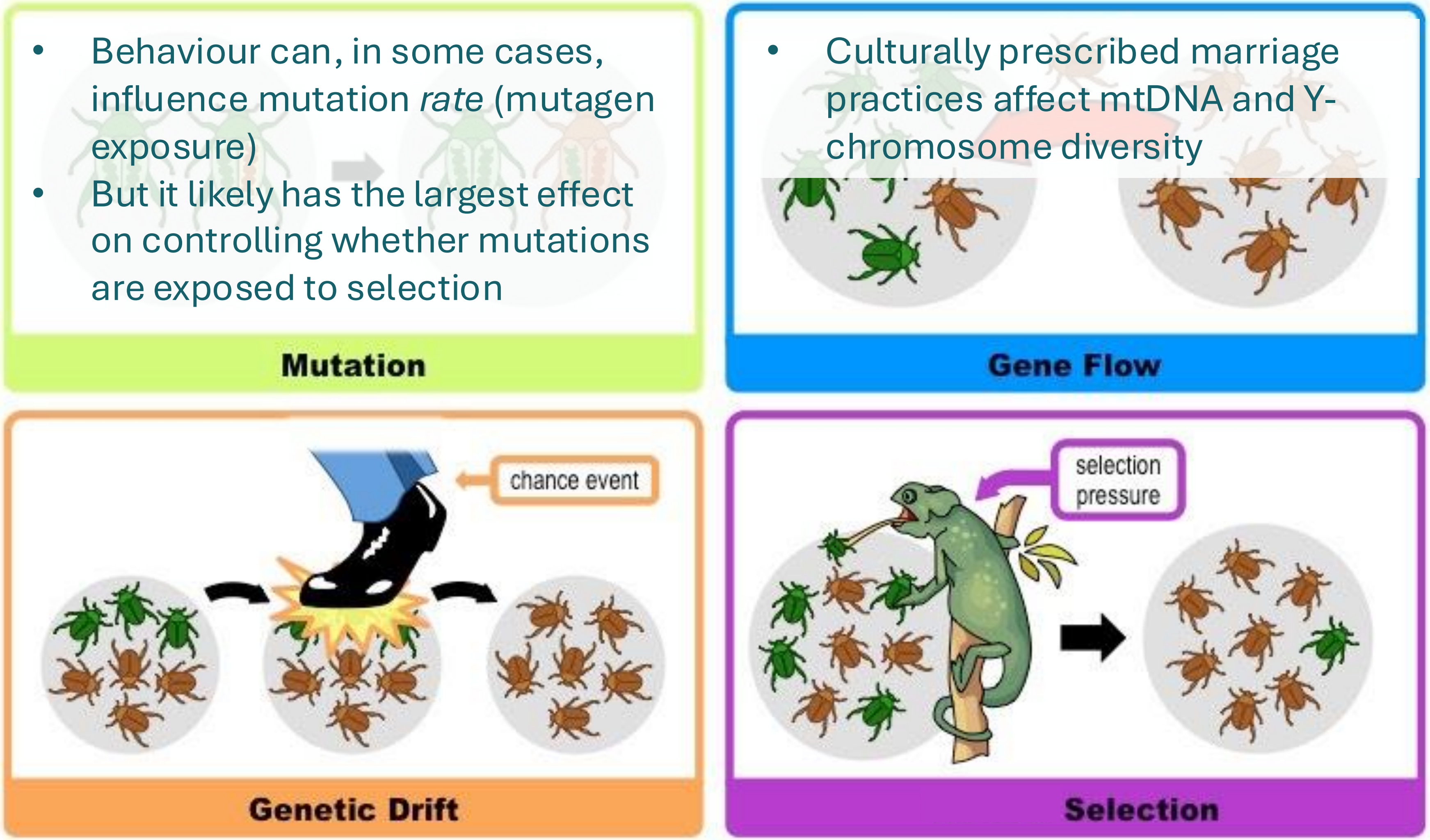
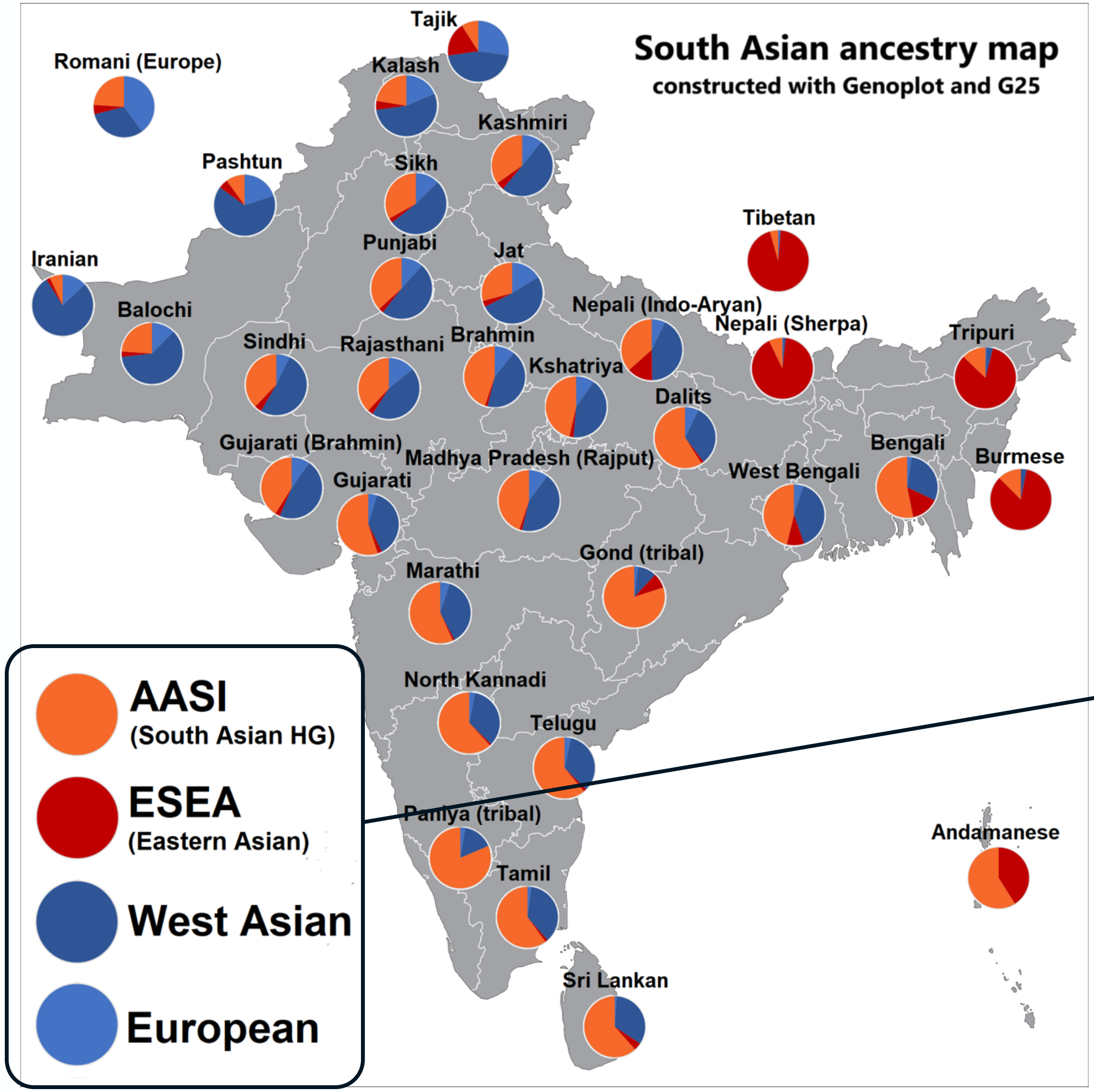


The curious case of India



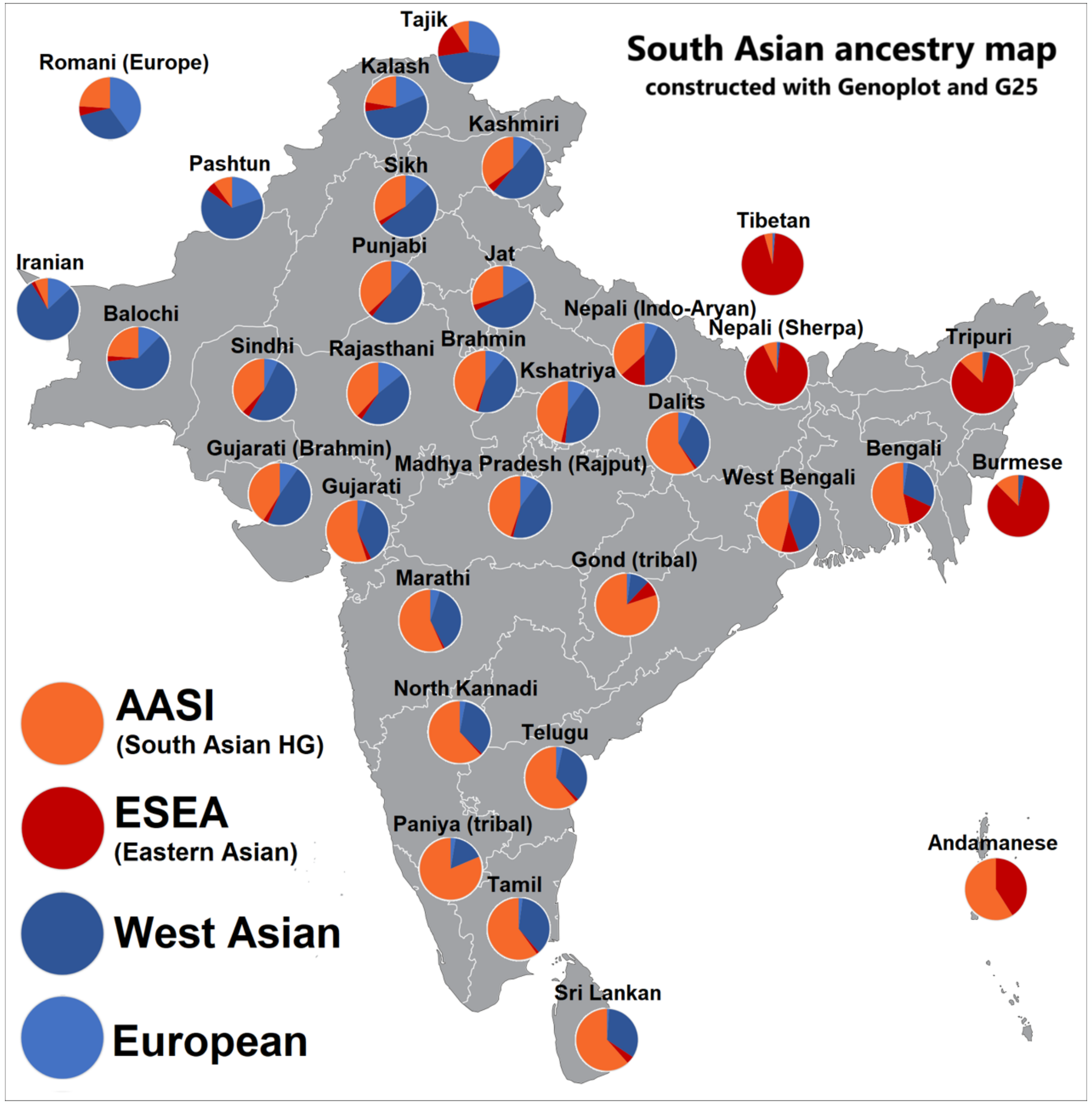
If mixing persists...

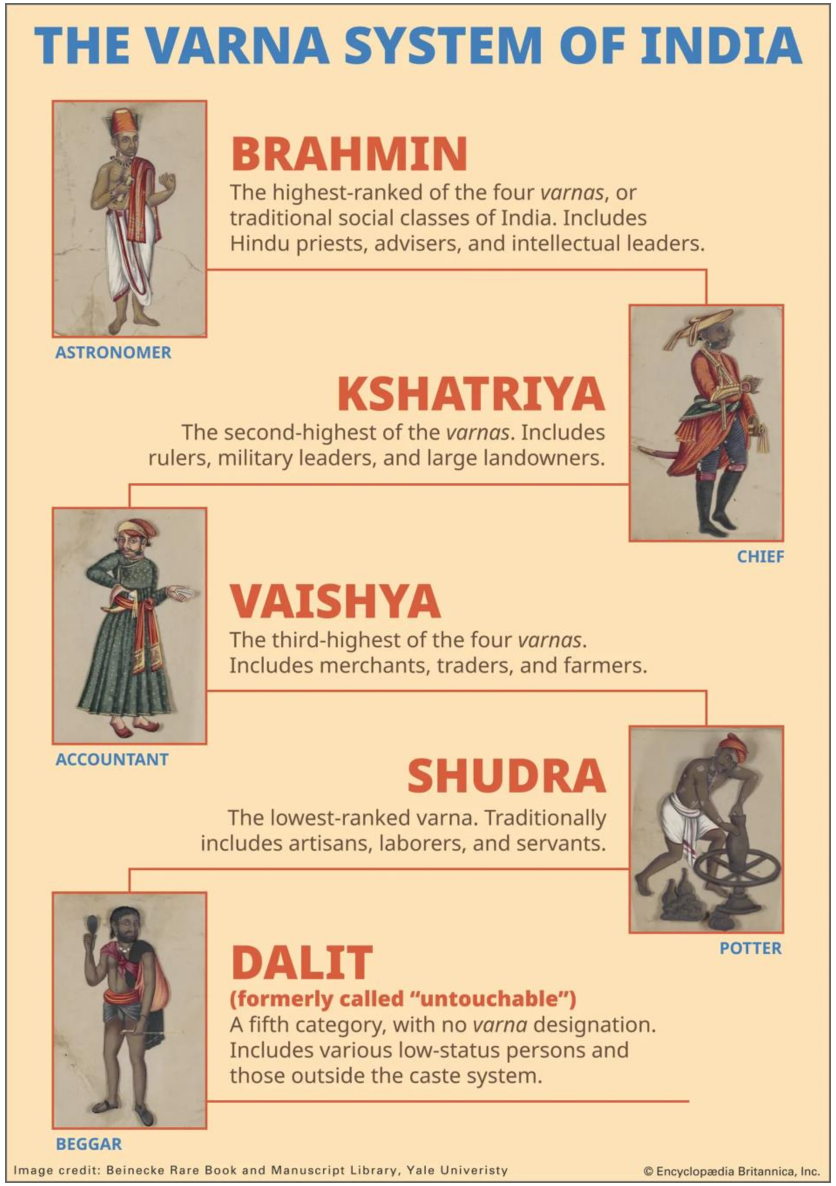
Amalgamation of all ancestries

Mixing stops 1900 years ago. Why?

Reich et al. (2009) *Nature*

The curious cas(t)e of India



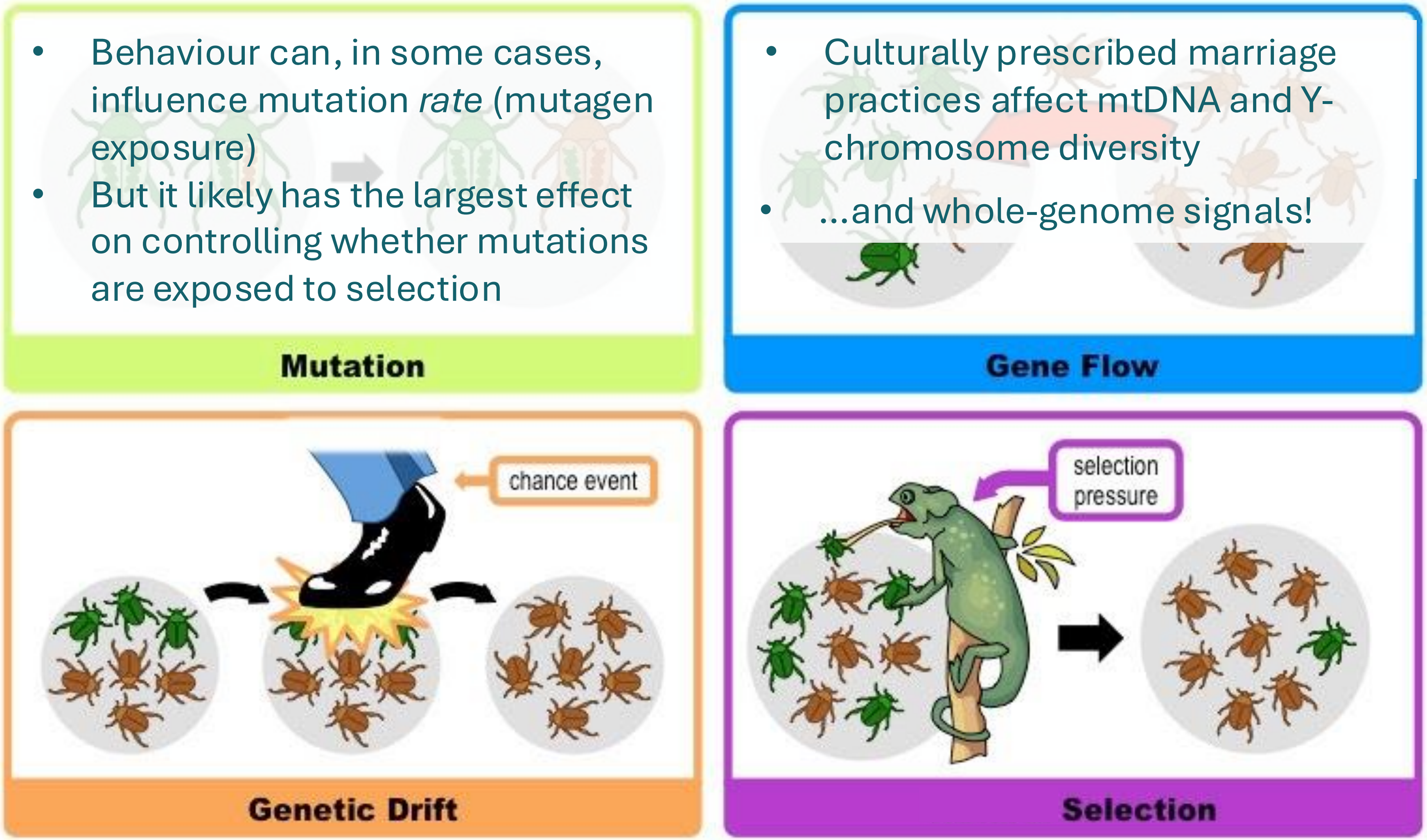


Reich et al. (2009) *Nature*

Which of these processes might be influenced by **culture** (or behaviour more generally)?

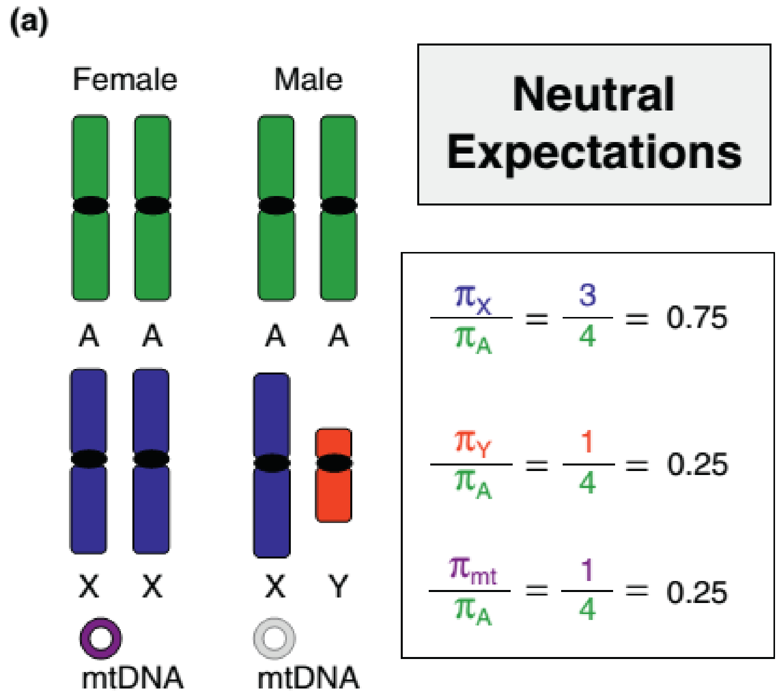
- Behaviour can, in some cases, influence mutation *rate* (mutagen exposure)
- But it likely has the largest effect on controlling whether mutations are exposed to selection

- Culturally prescribed marriage practices affect mtDNA and Y-chromosome diversity
- ...and whole-genome signals!



What is going on?

Key



Under random mating and neutrality, diversity of autosomes/ X/Y/mtDNA follow the **ploidy of transmissible copies** in the population

Neutral expectations:

- **N_e or diversity follows a 4:3:1:1 ratio across the autosomes, X, Y, and mtDNA respectively**
    - That is, autosomal diversity is expected to be 4x higher than mtDNA diversity; and Y and mtDNA are expected to be equally diverse

Sex-biased processes drive divergence from these expectations.

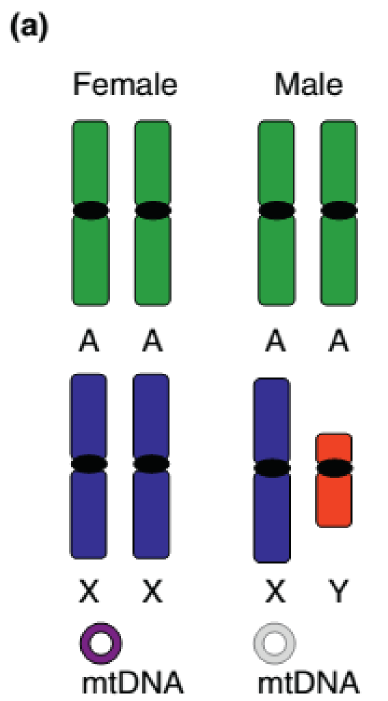
Reproductive skew

Marriage

- Unequal sex ratios (or large differences in effective population sizes between the sexes)
- Sex-biased migration and dispersal patterns
- Sex differences in generation time
- Sex-biased admixture and introgression

Webster and Wilson Sayres 2016 Curr Opin Genet Dev

What is going on?



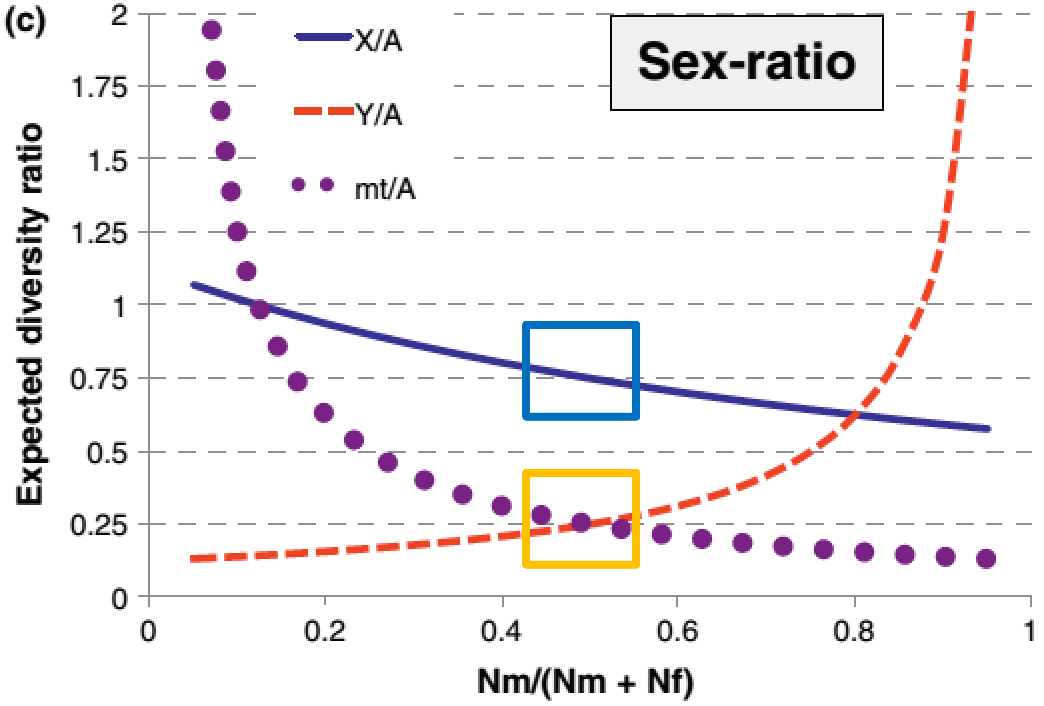
Neutral Expectations

$$\frac{\pi_X}{\pi_A} = \frac{3}{4} = \boxed{0.75}$$

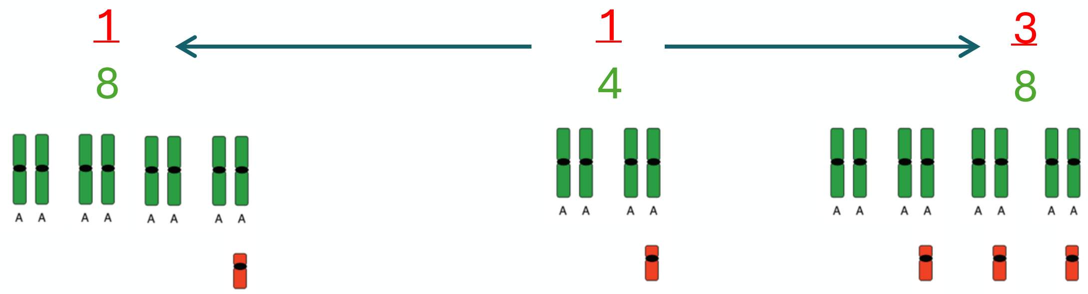
$$\frac{\pi_Y}{\pi_A} = \frac{1}{4} = 0.25$$

$$\frac{\pi_{mt}}{\pi_A} = \frac{1}{4} = 0.25$$

Key



% of the population that is male



Webster and Wilson Sayres 2016 Curr Opin Genet Dev

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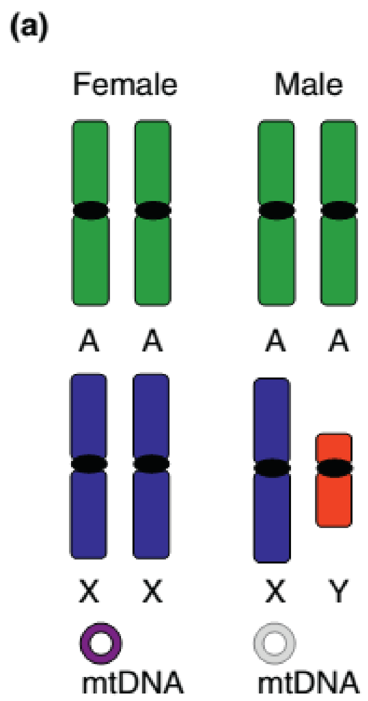

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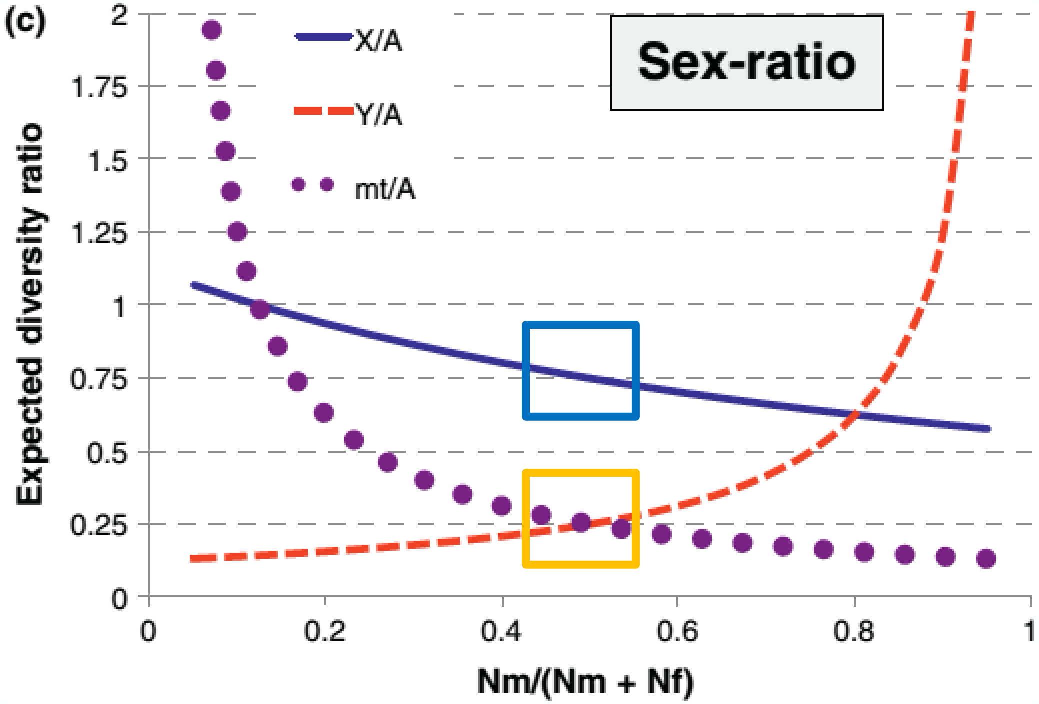

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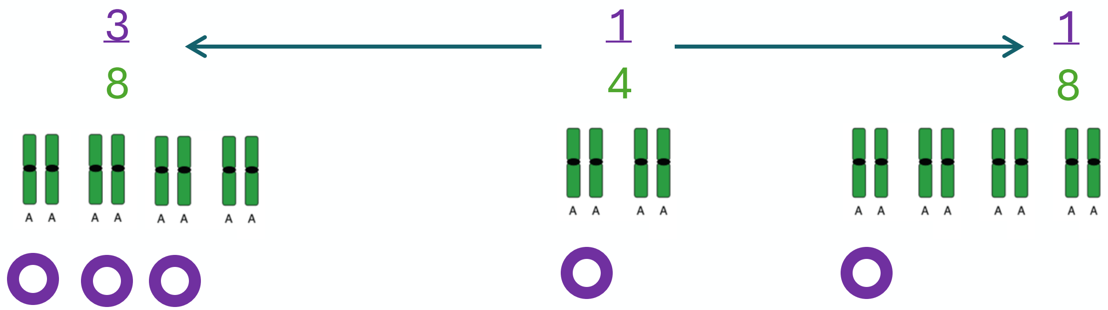
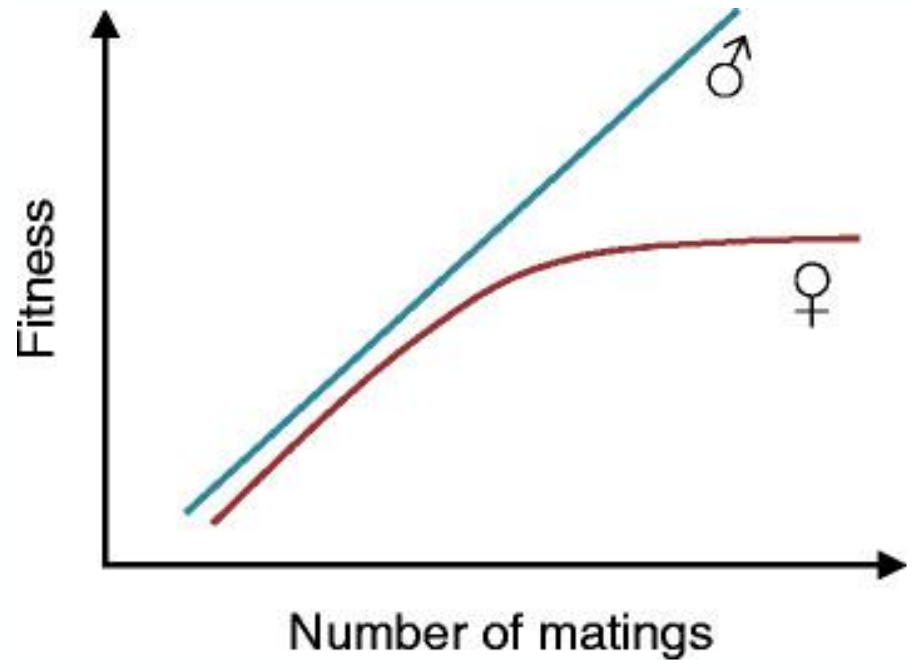
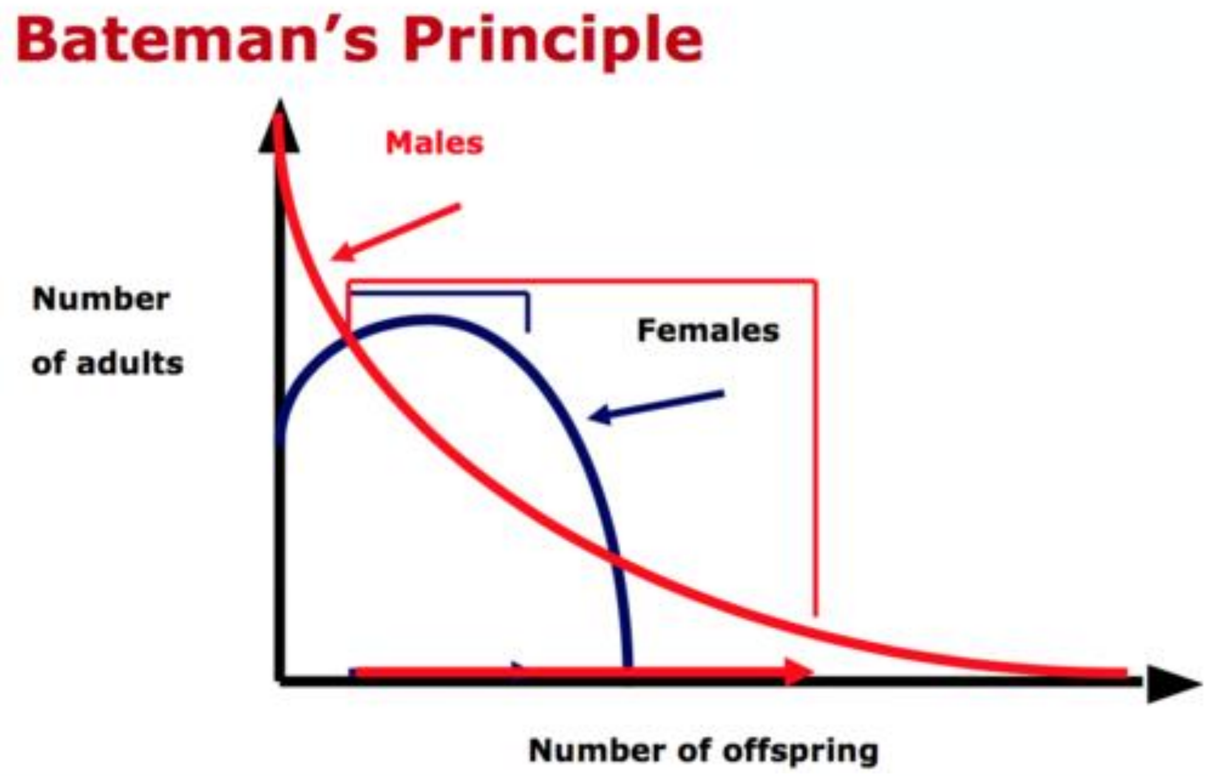


Webster and Wilson Sayres 2016 Curr Opin Genet Dev

Which sex can have a *very small* effective population size?

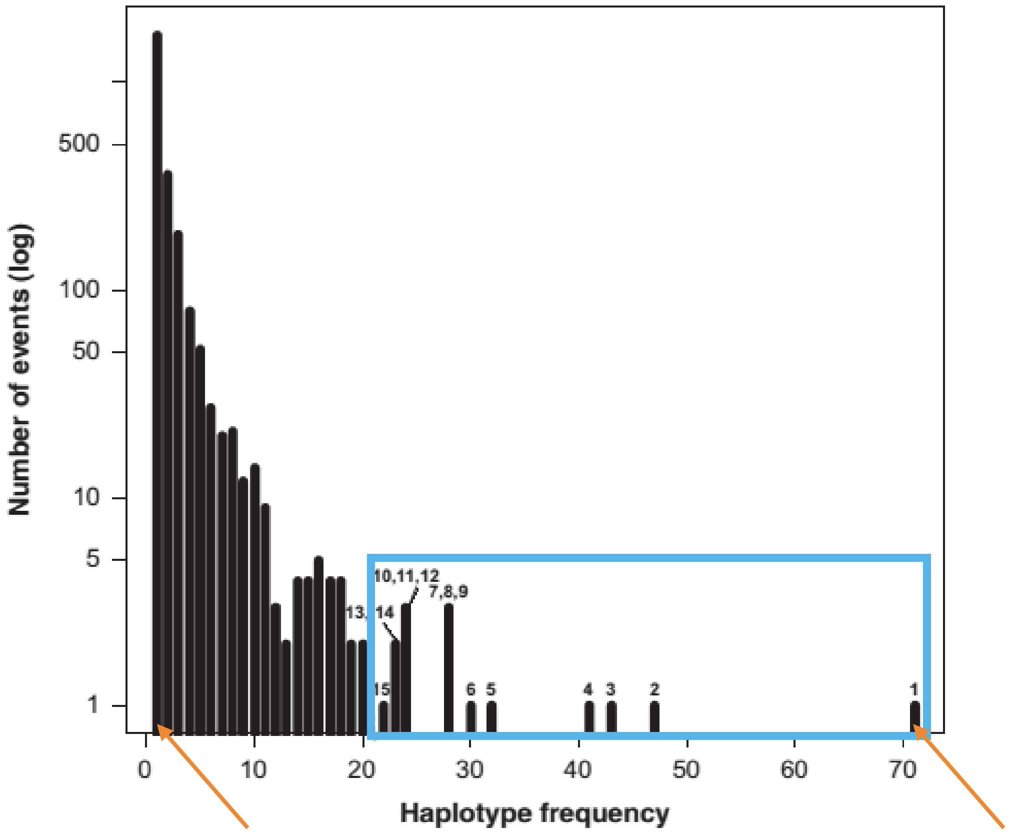




1) Asian Y-chromosome haplotypes

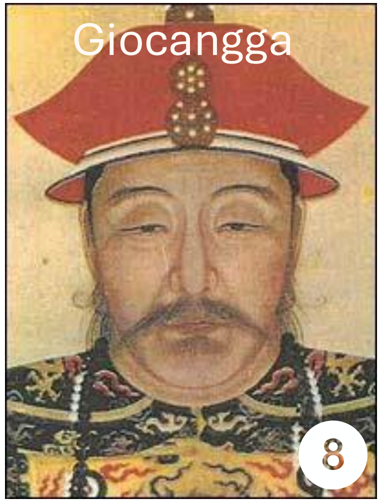
15 haplotypes out of 2552 represent ~40% of genetic data

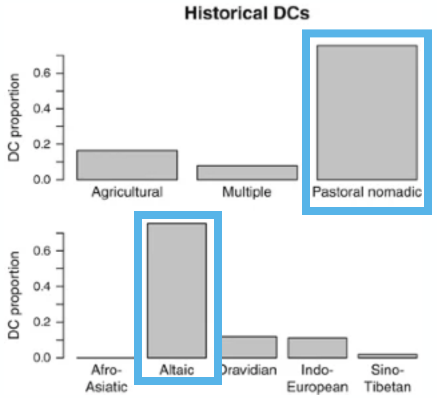
- Differential reproductive success



>500 haplotypes occur only once

1 haplotype occurs 70 times

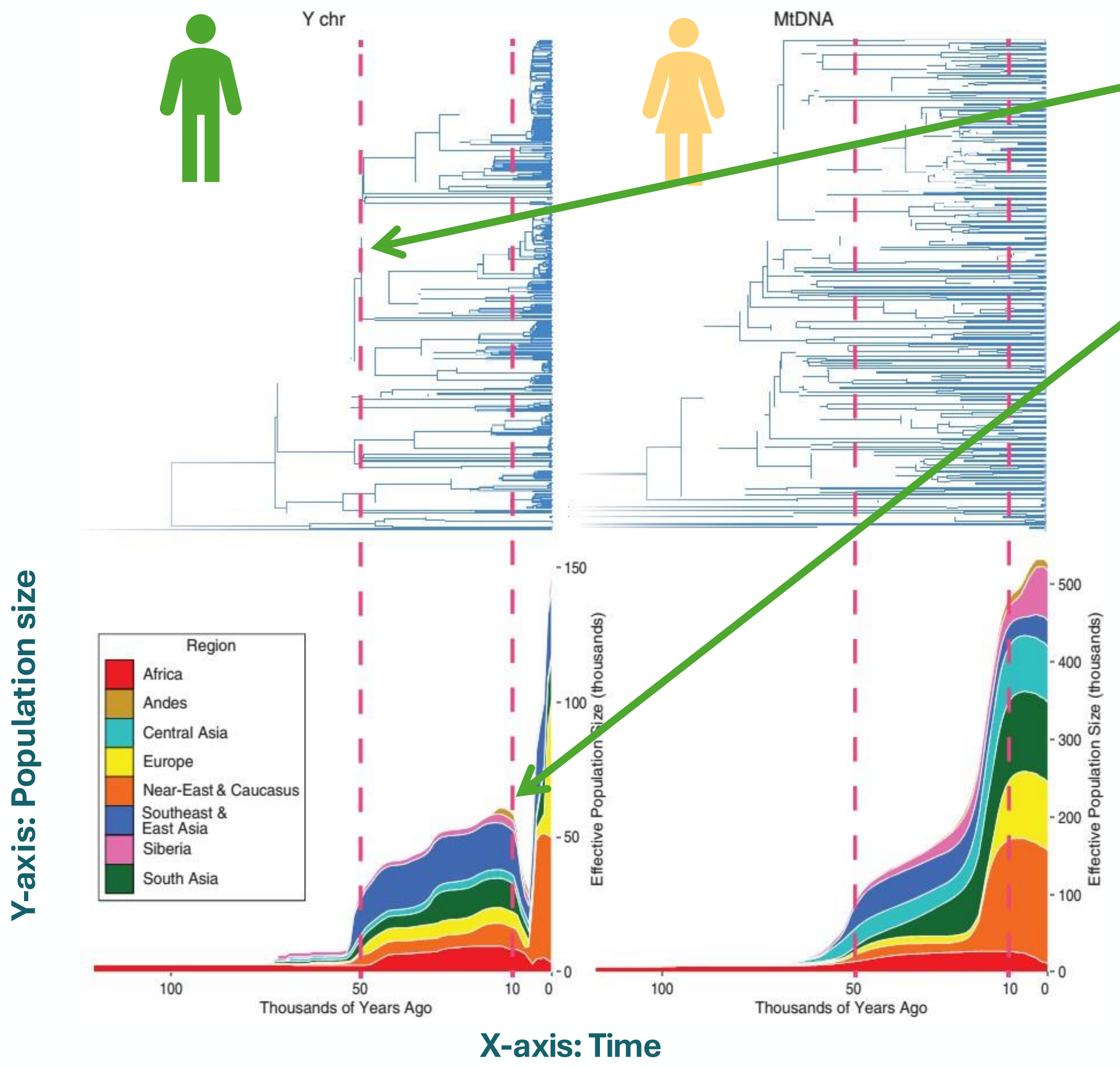




Pastoral nomadism, Altaic speaking

Balaresque et al. (2015) European Journal of Human Genetics

2) A male-specific population collapse in the Neolithic?



Coalescence of non-African populations

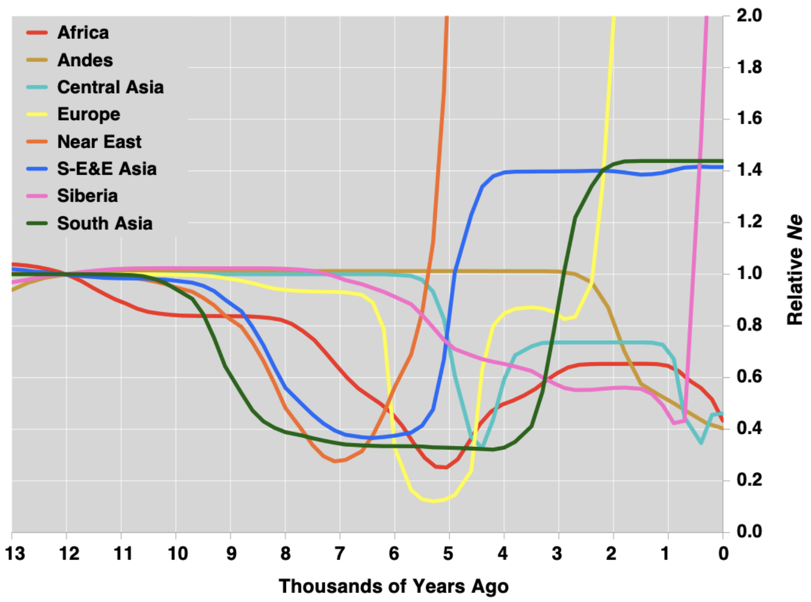
Archaeological evidence for OOA **50 kya**

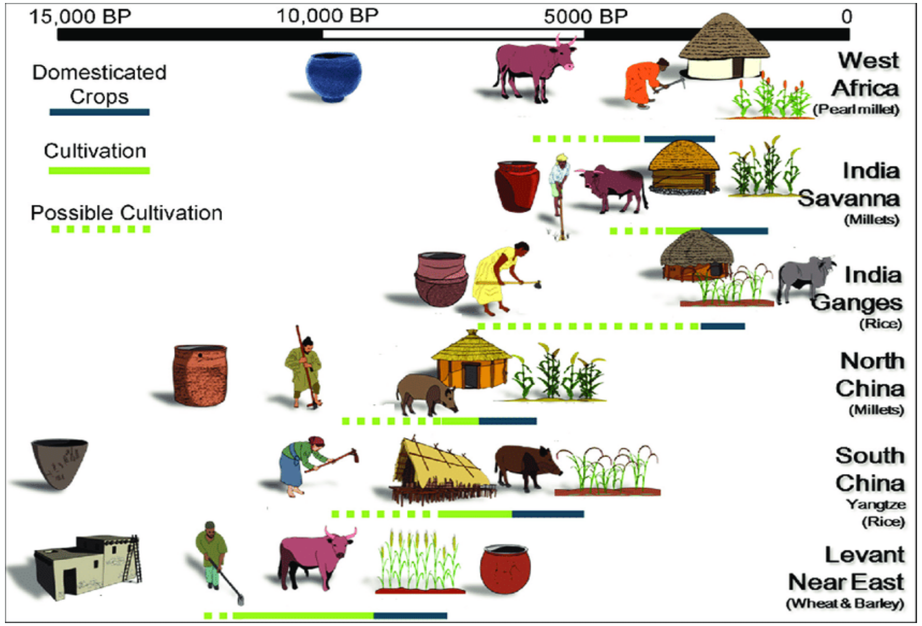
10 kya: steep drop in male effective population size

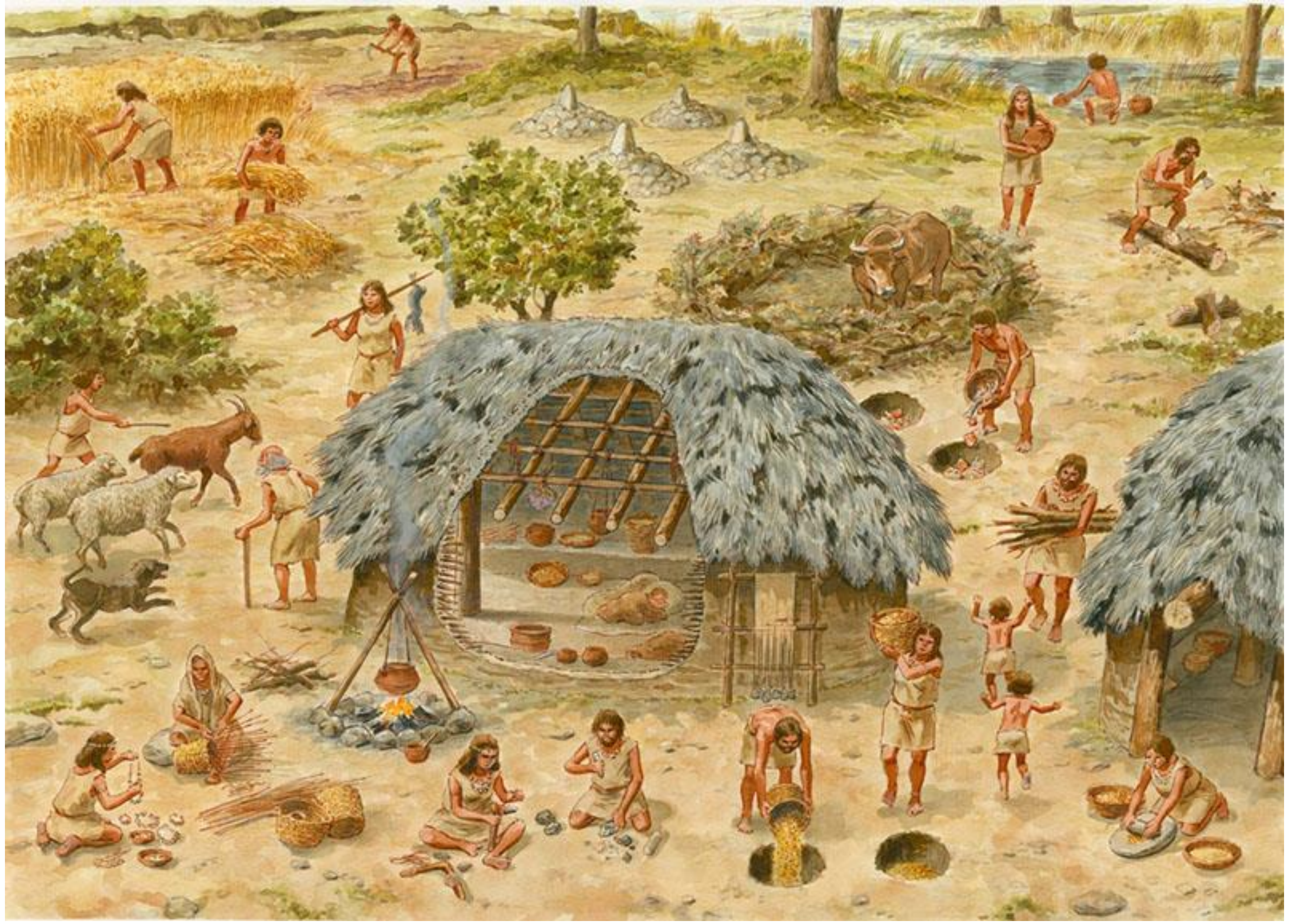
- Selection on Y-chromosome?
- Changes in variance of reproductive success?

Karmin et al 2015 Genom Res

A male-specific population collapse in the Neolithic?







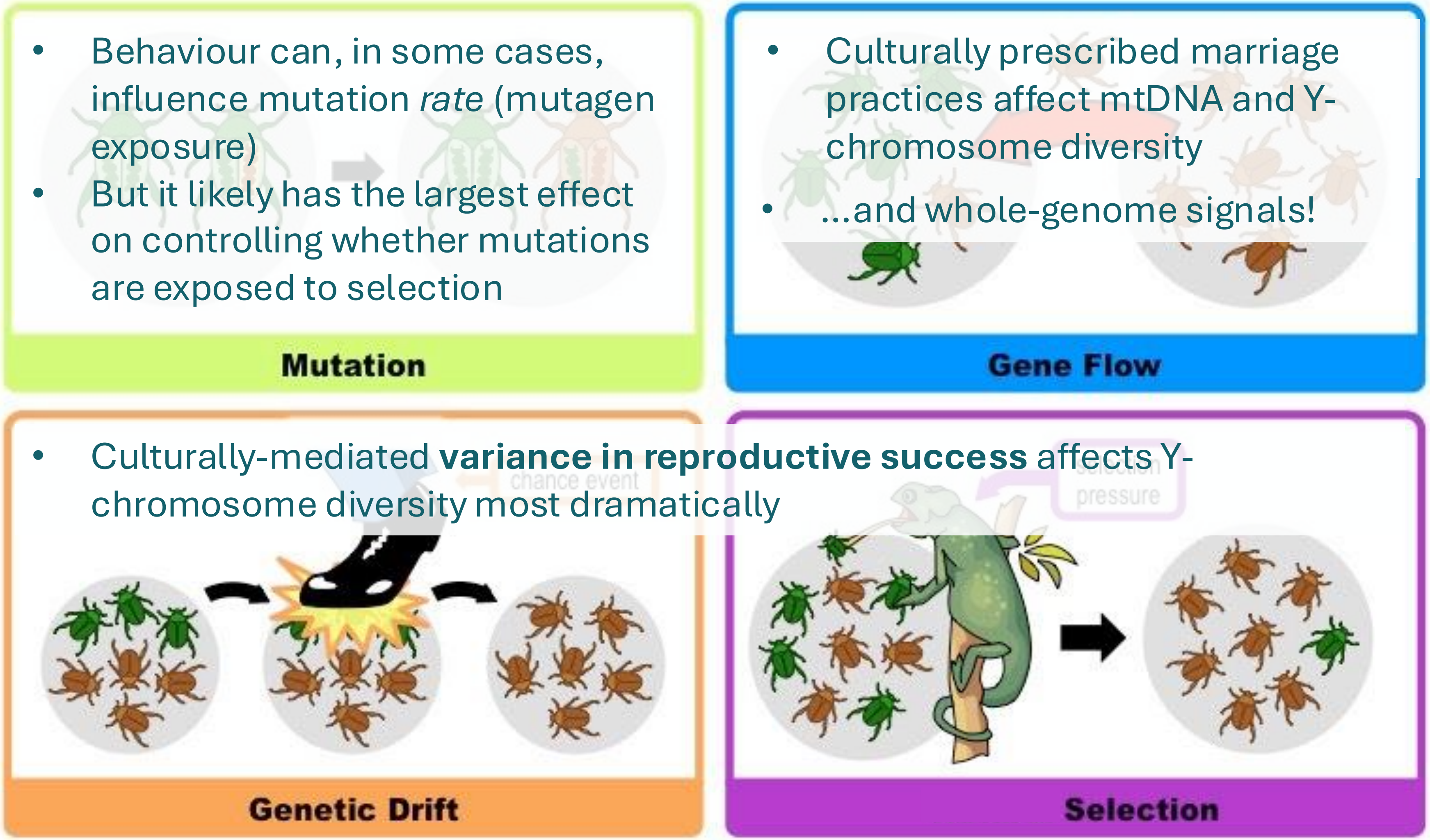
Karmin et al 2015 Genom Res

Which of these processes might be influenced by **culture** (or behaviour more generally)?

- Behaviour can, in some cases, influence mutation *rate* (mutagen exposure)
- But it likely has the largest effect on controlling whether mutations are exposed to selection

- Culturally prescribed marriage practices affect mtDNA and Y-chromosome diversity
- ...and whole-genome signals!

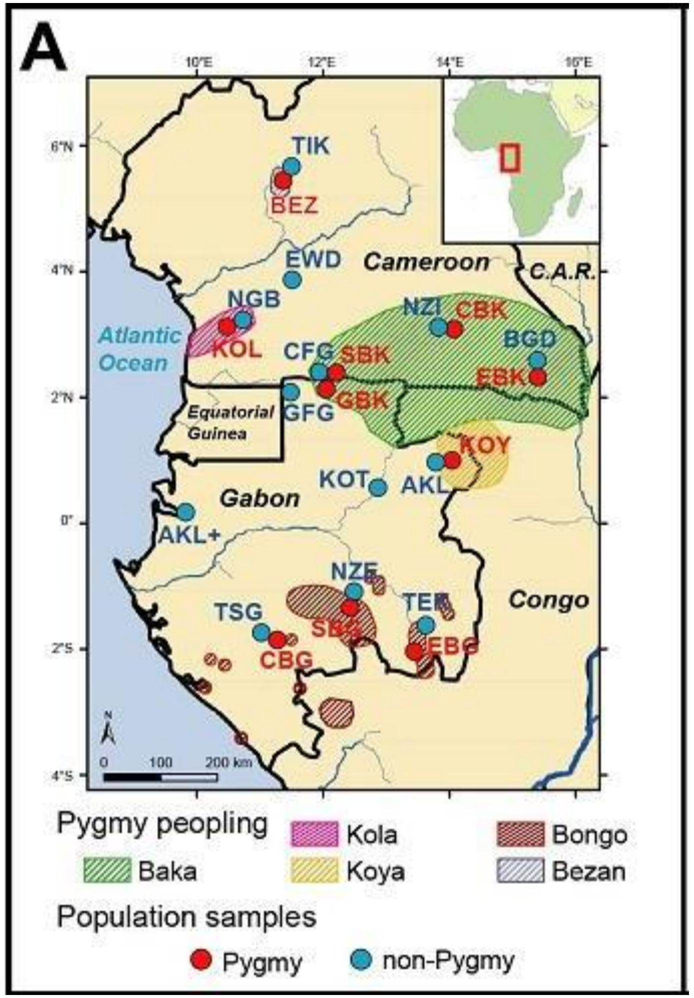
- Culturally-mediated **variance in reproductive success** affects Y-chromosome diversity most dramatically



Intro
Mutation
Gene flow
D+S

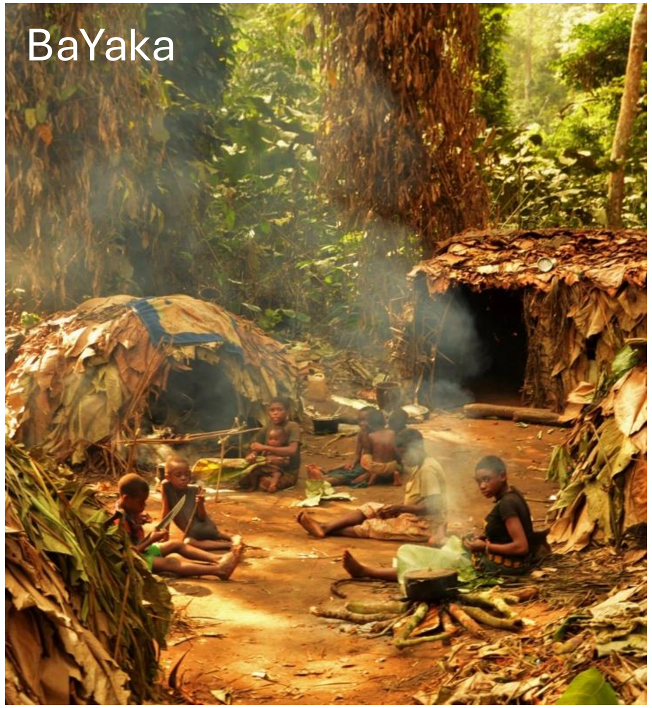
Patrilocality and variance in male reproductive success are not uncorrelated!

Patrilocality and variance in male reproductive success are not uncorrelated… and these are not the only cultural drivers of note!



Central African Pygmy versus non-Pygmy groups

Terminology note – 'Pygmy' is a group-defined phenotype (average adult male height <150cm), commonly used by anthropologists who know the mainly Central African traditionally rainforest hunter-gatherer groups well. But, if referring to a group of people and not the phenotype and the behavioural and evolutionary context of that phenotype, then certainly the community group name is better!



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Verdu et al 2013 MBE

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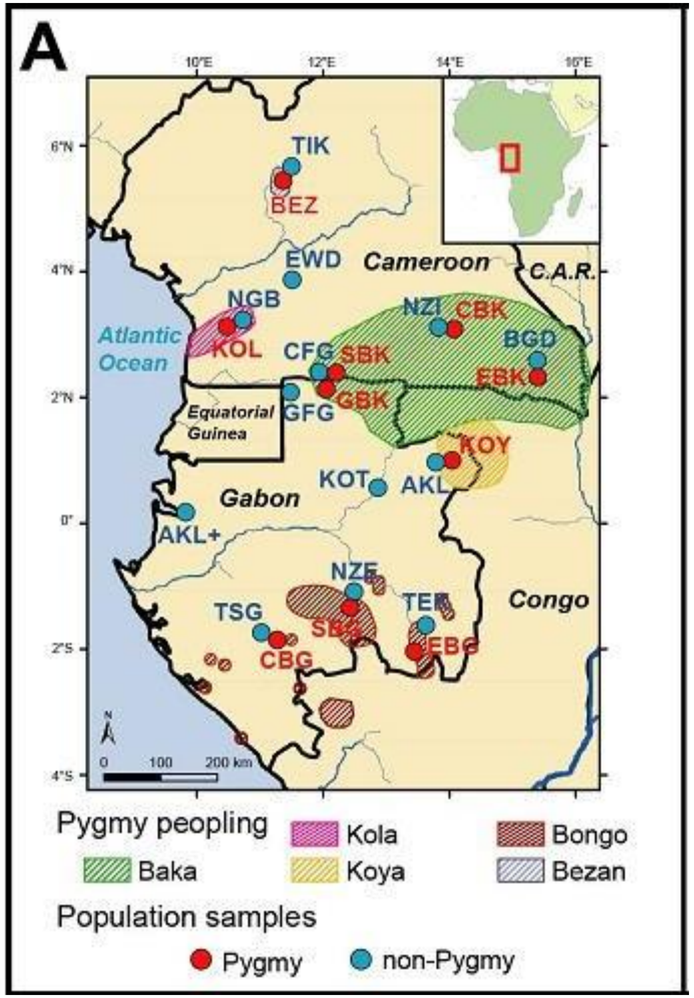

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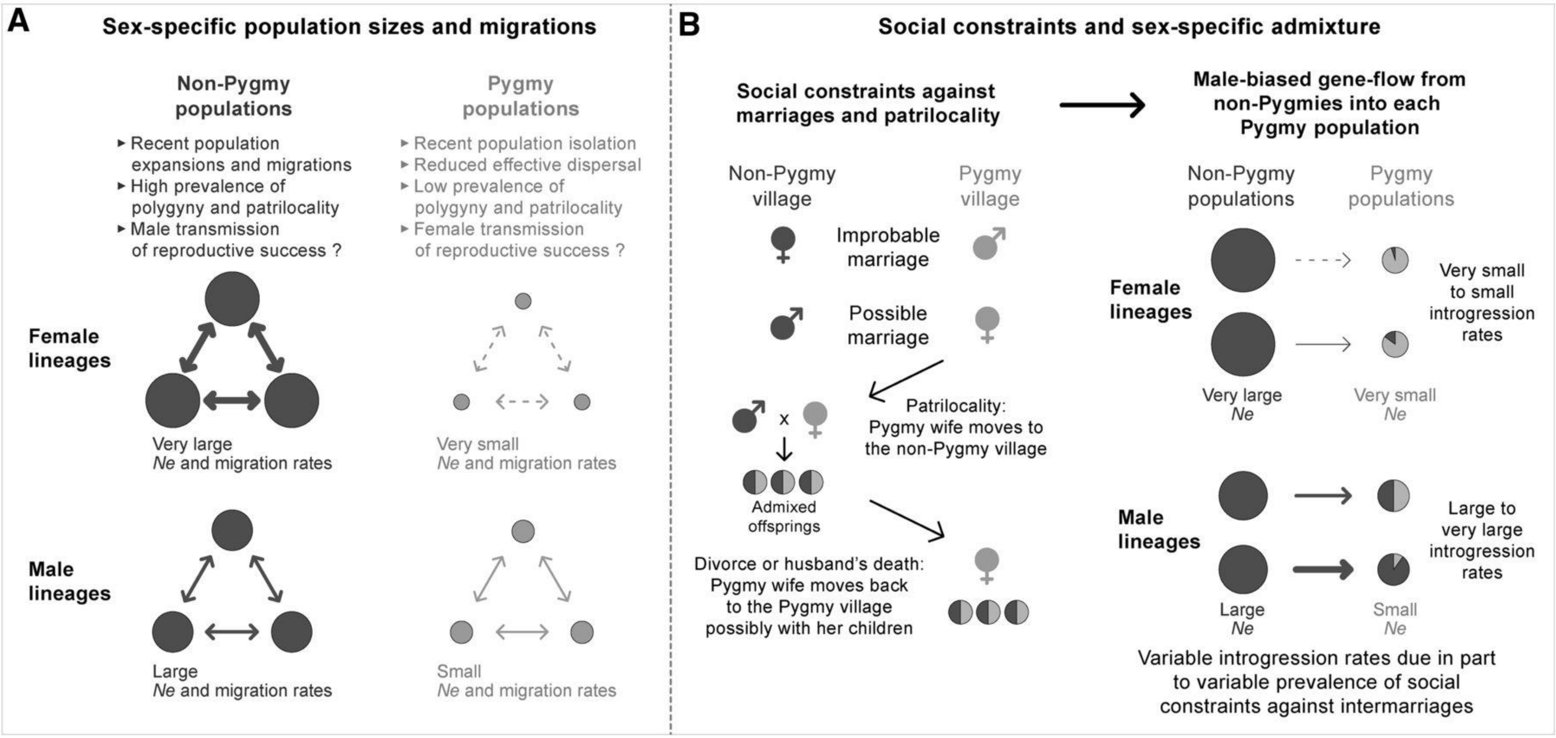

NP: Expect reduced Y-chromosome diversity

P: Expect reduced mtDNA diversity

Verdu et al 2013 MBE

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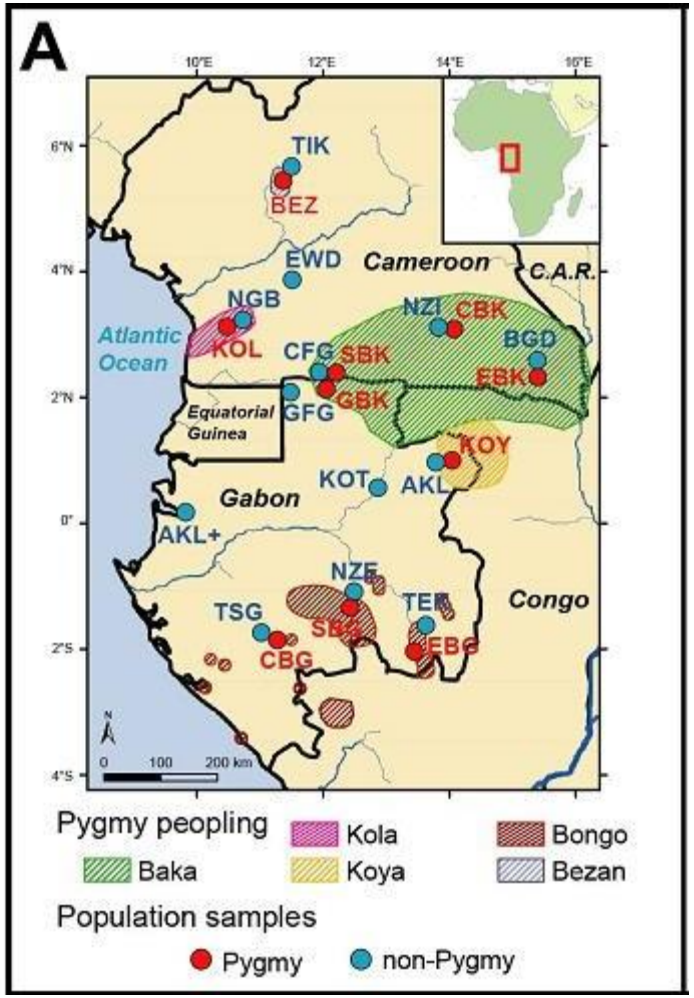

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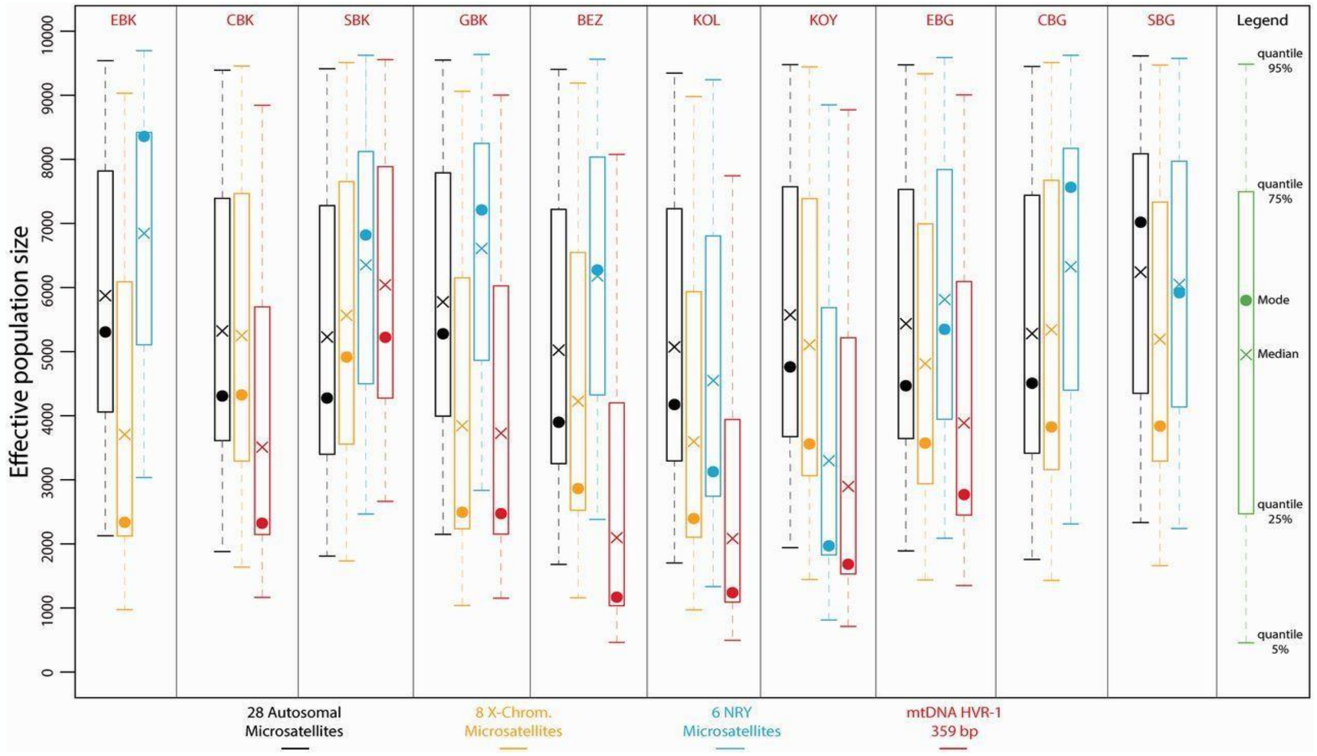

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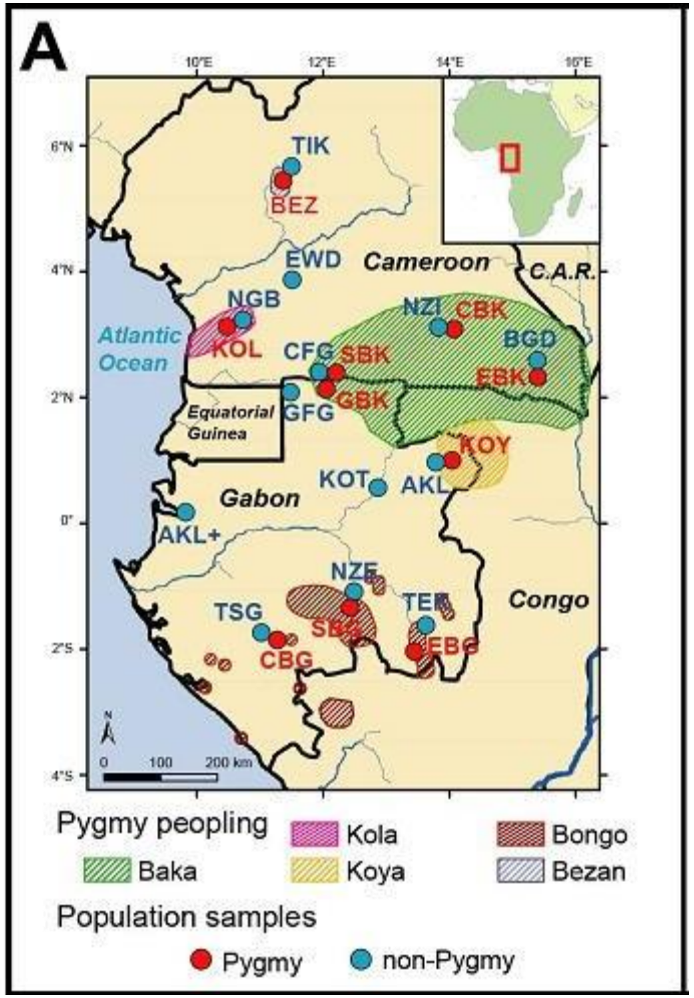

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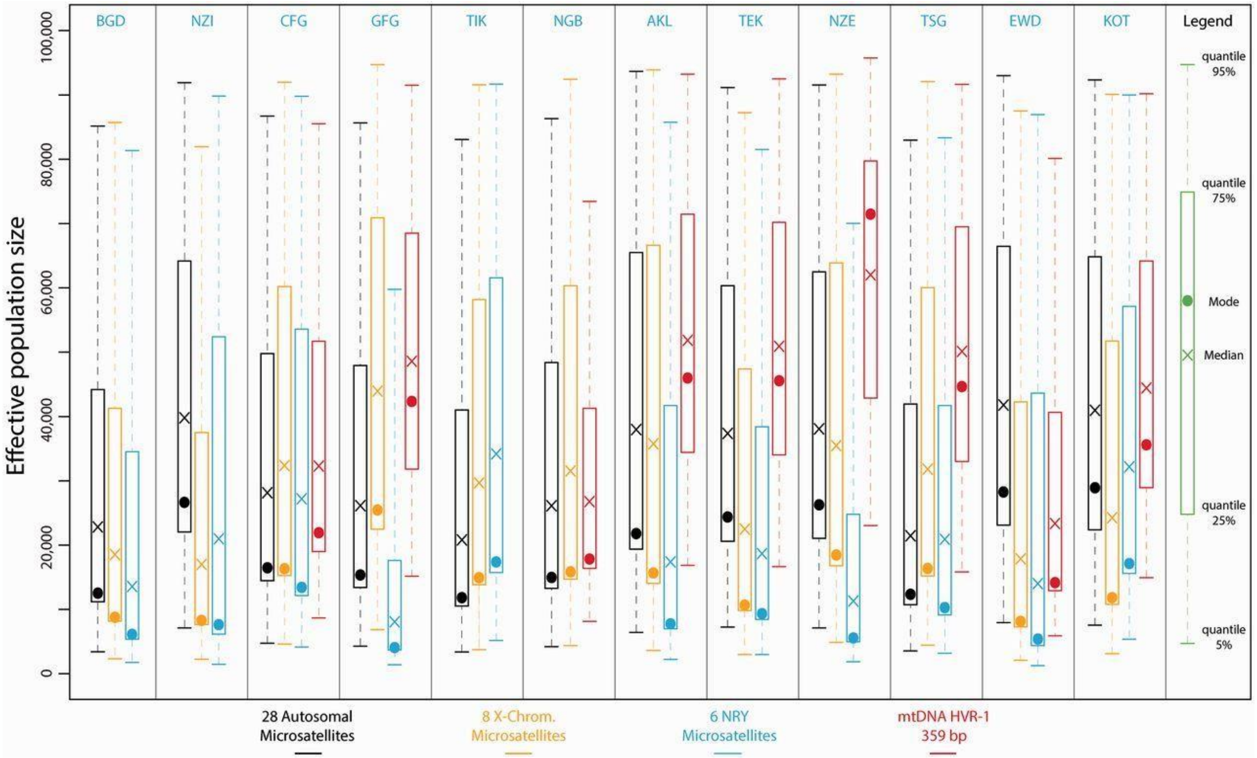

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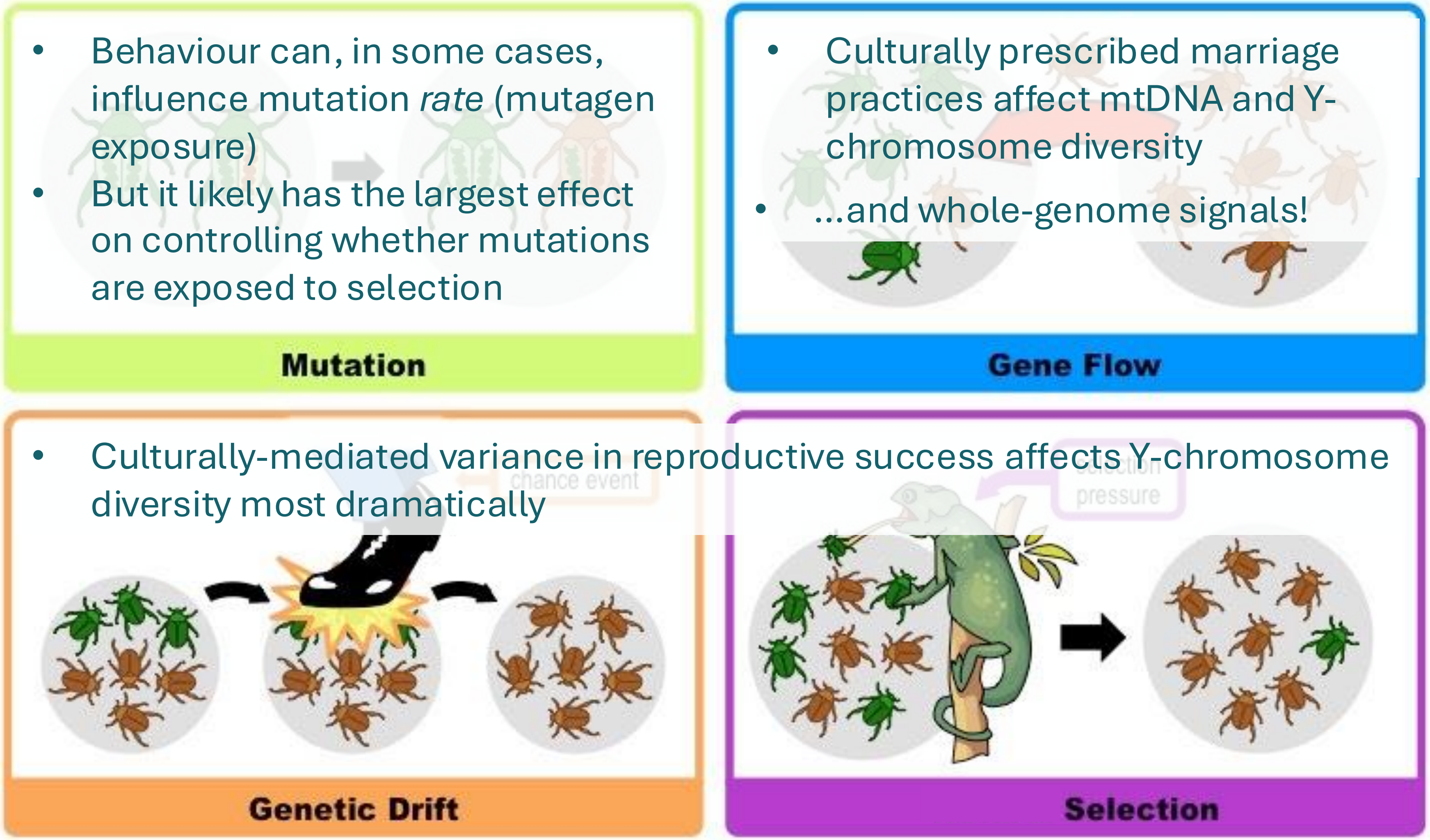
Verdu et al 2013 MBE

Summary



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Genetic Drift / Selection

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